

ac Steady-State Circuit Analysis

AI.1 | PHASOR REPRESENTATION OF SINE WAVES

The sinusoidal waveform of Fig. AI.1 can be represented as

$$a = \sqrt{2} A \sin(\omega t + \phi)$$

where

$\sqrt{2}A$ = maximum amplitude of sine wave

A = rms value of sine wave

$\omega = 2\pi f$ = frequency in rad/s

f = frequency in Hz (cycles/s)

ϕ = phase angle

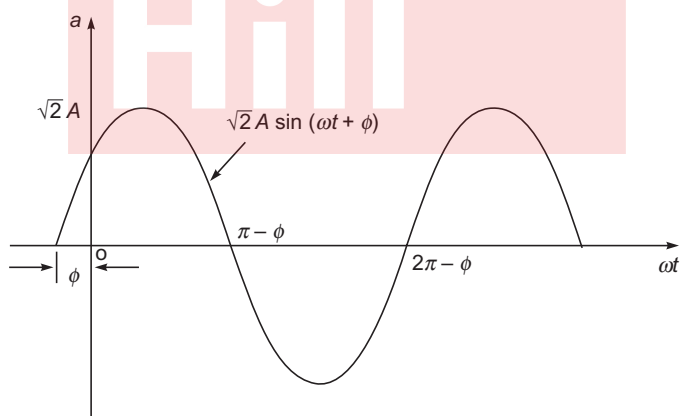


Fig. AI.1

It can also be written as

$$a = \sqrt{2} A \sin(\omega t + \phi) = \text{Im}[\sqrt{2} A e^{j\phi} e^{j\omega t}] \quad (\text{AI.1a})$$

$$= \text{Im} \left[\sqrt{2} \bar{A} e^{j\omega t} \right] \quad (\text{AI.1b})$$

where

$$\bar{A} = A e^{j\phi} = A \angle \phi \quad (\text{AI.2})$$

According to Eq. (AI.1b) a sine wave can be represented by the phasor $\bar{A}$ as defined in Eq. (AI.2) where A is the magnitude of the phasor representing the rms value of the sine wave and ϕ is its phase angle. By performing the imaginary operation on $\sqrt{2} \bar{A} e^{j\omega t}$ as in Eq. (AI.1b), the instantaneous value of the sine wave represented by the phasor $\bar{A}$ can be found. Another powerful way is the graphical representation of a phasor as shown in Fig. AI.2 wherein the phasor $\bar{A}$ is represented by a directed line of magnitude A located at an angle ϕ from the reference axis. Anti-clockwise rotation at ω rad/s is imparted to the phasor $\bar{A}$ by $e^{j\omega t}$ in Eq. (AI.1b). It easily follows from the graphical representation that the projection of the phasor on the y-axis at time t multiplied by $\sqrt{2}$ gives the instantaneous value of the sine wave represented by the phasor*.

The phasor concept is very useful in ac steady-state circuit analysis.

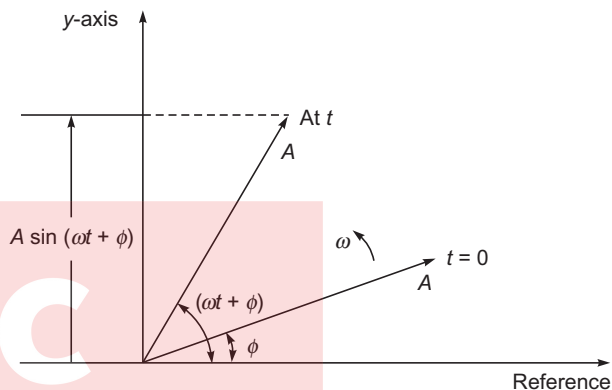


Fig. AI.2 Phasor representation of sine wave

AI.2 | ADDITION OF PHASORS

Addition (or subtraction) of two or more sine waves of the same frequency can be easily carried out by means of phasor representation. For example,

$$\sqrt{2} A_1 \sin (\omega t + \phi_1) + \sqrt{2} A_2 \sin (\omega t + \phi_2) = \sqrt{2} A \sin (\omega t + \phi) \quad (\text{AI.3})$$

The two sine waves in Eq. (AI.3) are represented by phasors

$$\bar{A}_1 = A_1 e^{j\phi_1}$$

$$\bar{A}_2 = A_2 e^{j\phi_2}$$

Since they are of the same frequency, they are relatively stationary with respect to each other as shown in the phasor diagram of Fig. AI.3. Therefore, in the process of addition, the rotation of phasors has no significance; it needs only to be brought into picture when the instantaneous values of a sine wave are required. It is also to be observed that the phasor method is useful in dealing with sine waves of the same frequency†.

In its phasor form Eq. (AI.3) is written as

$$\bar{A}_1 + \bar{A}_2 = \bar{A} = A e^{j\phi} \quad (\text{AI.4})$$

The corresponding graphical phasor addition is shown in Fig. AI.3. Phasor subtraction is shown in Fig. AI.4.

* Phasor representing sine wave in time is also called a *time phasor*.

† The phasor concept can be extended to addition (subtraction) of two sine waves of different frequencies wherein the two phasor move relative to each other and the resultant phasor at any instant yields the instantaneous rms value and phase angle of the sum (see Sec. 3.13).

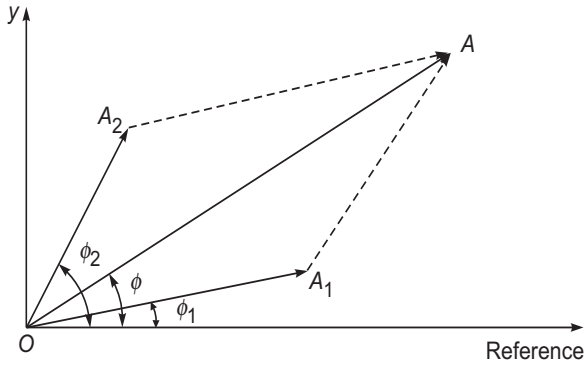


Fig. AI.3

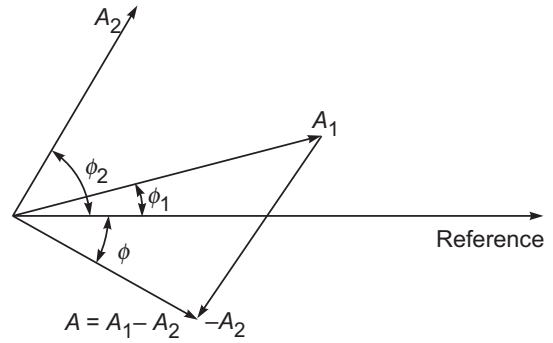


Fig. AI.4

AI.3 | AC STEADY-STATE ELEMENTAL BEHAVIOUR

A circuit excited with ac after the transients have died out, reaches steady-state wherein voltages and currents at all points in the circuit are sinusoidal provided the circuit is composed of linear elements. Therefore, ac steady-state behaviour of linear lumped elements—resistance, inductance and capacitance—can be modelled in terms of phasor voltages and currents.

Resistance

For the resistance element shown in Fig. AI.5(a),

$$\bar{V}_R = R\bar{I}_R \quad (\text{AI.5})$$

wherein the voltage across resistance is *in phase* with current though it as shown by the phasor diagram of Fig. AI.5(b). Equation (AI.5) can also be written as

$$\bar{I}_R = G\bar{V}_R \quad (\text{AI.6})$$

where

$$G = \frac{1}{R} = \text{conductance}$$

Inductance

For the inductance element (L) of Fig. AI.6(a)

$$\bar{V}_L = jX_L\bar{I}_L \quad (\text{AI.7})$$

where

$$X_L = \omega L = 2\pi fL = \text{inductive reactance} \quad (\text{AI.8})$$

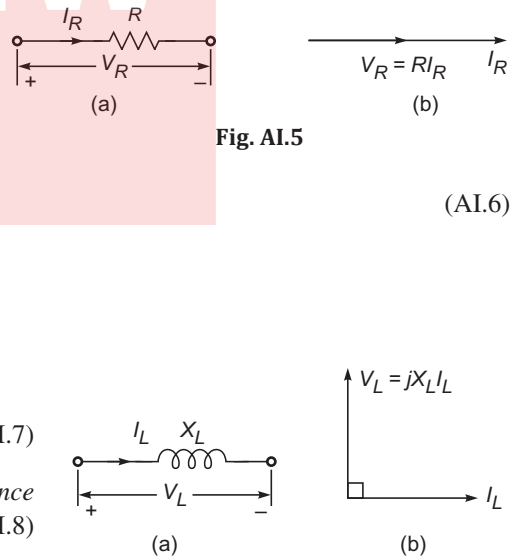


Fig. AI.6

The phasor diagram corresponding to the elemental behaviour of Eq. (AI.7) is drawn in Fig. AI.6(b) wherein the voltage across inductance and current through it are in phase quadrature with the current lagging the voltage by 90° .

Equation (AI.7) can also be written as

$$\bar{I}_L = -jB_L \bar{V}_L \quad (\text{AI.9})$$

where

$$B_L = \frac{1}{X_L} = \frac{1}{\omega L} = \text{inductive susceptance} \quad (\text{AI.10})$$

Capacitance

For the capacitance element (C) of Fig. AI.7(a)

$$\bar{V}_C = -jX_C \bar{I}_C \quad (\text{AI.11})$$

where

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C} = \text{capacitive reactance} \quad (\text{AI.12})$$

Corresponding to Eq. (AI.11), the phasor diagram is drawn in Fig. AI.7(b), wherein current leads the voltage by 90° . Equation (AI.11) can also be written as

$$\bar{I}_C = jB_C \bar{V}_C \quad (\text{AI.13})$$

where

$$B_C = \frac{1}{X_C} = \omega C = \text{capacitive susceptance}$$

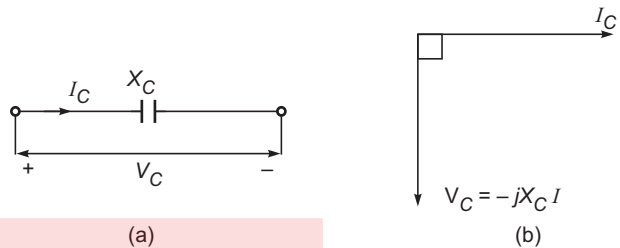


Fig. AI.7

AI.4 | SERIES RL CIRCUIT

Applying Kirchhoff's voltage law in phasor form to the series RL circuit of Fig. AI.8(a).

$$\begin{aligned} \bar{V} &= \bar{V}_R + \bar{V}_L \\ &= R\bar{I} + jX_L \bar{I} = (R + jX_L)\bar{I} \end{aligned} \quad (\text{AI.14})$$

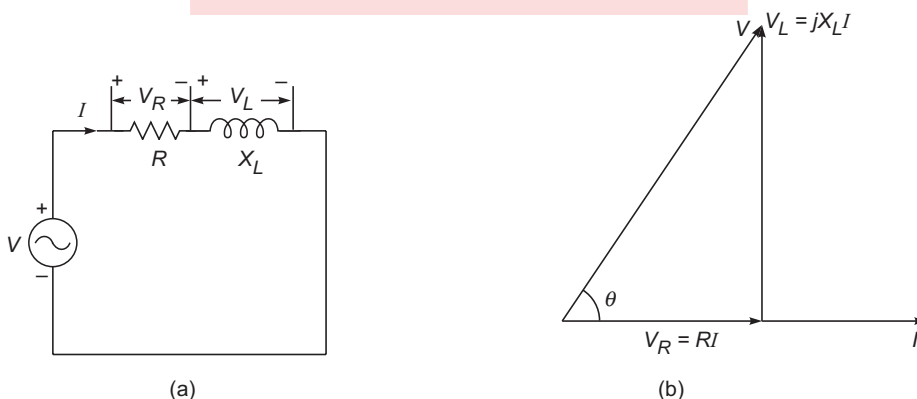


Fig. AI.8

By use of elemental relations described earlier, the phasor diagram corresponding to Eq. (AI.13) is drawn in Fig. AI.8(b) from which it is observed that the current lags voltage by

$$\theta = \tan^{-1} \frac{X_L}{R} \quad (\text{AI.15})$$

Equation (AI.14) can be written as

$$\bar{I} = \frac{\bar{V}}{R + jX_L} \quad (\text{AI.16})$$

Impedance of a circuit is defined as

$$\begin{aligned} \bar{Z} &= (R + jX_L) = \text{impedance (complex number)} \\ &= Z \angle \theta \end{aligned} \quad (\text{AI.17})$$

where

$$\theta = \tan^{-1} \frac{X_L}{R}$$

which in fact is the angle by which the current lags the voltage in the circuit. The impedance of the circuit is represented as a complex number in Fig. AI.9. It is also observed that impedance is not a phasor.

Equation (AI.16) can also be written as

$$\bar{I} = \frac{\bar{V}}{\bar{Z}} = \frac{V}{Z} \angle -\theta \quad (\text{with } \bar{V} = V \angle 0^\circ) \quad (\text{AI.18})$$

which is nothing but Ohm's law in phasor form. Also

$$\bar{I} = \bar{Y} \bar{V} \quad (\text{AI.19})$$

where

$$\bar{Y} = \frac{1}{\bar{Z}} = \text{admittance (complex number)} \quad (\text{AI.20a})$$

$$= G - jB \quad (\text{AI.20b})$$

where

G = circuit conductance

B = circuit susceptance

It easily follows that in this circuit

$$G = \frac{R}{R^2 + X_L^2} \quad (\text{AI.21})$$

$$B = \frac{X_L}{R^2 + X_L^2} \quad (\text{AI.22})$$

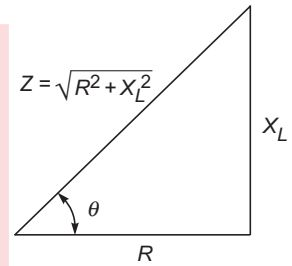


Fig. AI.9 Impedance triangle

AI.5 | PARALLEL RL CIRCUIT

Consider the parallel RL circuit of Fig. AI.10(a) excited by voltage $\bar{V}$. The two circuit elements are described by

$$\bar{I}_R = \frac{\bar{V}}{R} = G\bar{V}; \quad \bar{I}_L = \frac{\bar{V}}{jX_L} = -jB_L\bar{V}$$

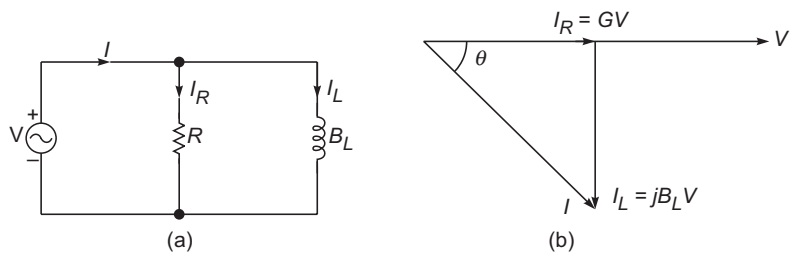


Fig. AI.10 Parallel RL circuit

Applying Kirchhoff's current law

$$\begin{aligned}\bar{I} &= \bar{I}_R + \bar{I}_L \\ &= (G - jB_L)\bar{V} = \bar{Y}\bar{V}\end{aligned}\quad (\text{AI.23})$$

where

$$\bar{Y} = G - jB_L$$

The phasor diagram as per Eq. (AI.23) is drawn in Fig. AI.10(b) wherein

$$\theta = \tan^{-1} \frac{B_L}{G} = \text{angle by which current lags the applied voltage}$$

It is now easily recognized from Eq. (AI.20b) that the parallel equivalent of the series circuit of Fig. (AI.8) is as shown in Fig. (AI.11) wherein G and B are given by Eqs (AI.21) and (AI.22).

It is suggested that the reader find the equivalent series form for the parallel RL circuit of Fig. AI.10 by use of Eq. (AI.23).

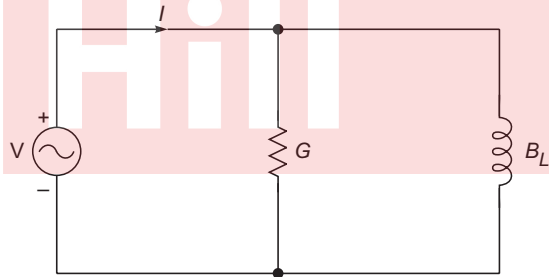


Fig. AI.11 Parallel equivalent of the series RL circuit of Fig. AI.8

AI.6 | IMPEDANCES/ADMITTANCES IN SERIES/PARALLEL

For impedances in series

$$\bar{Z} = \bar{Z}_1 + \bar{Z}_2 + \dots \quad (\text{AI.24})$$

or

$$\frac{1}{\bar{Y}} = \frac{1}{\bar{Y}_1} + \frac{1}{\bar{Y}_2} + \dots \quad (\text{AI.25})$$

For admittances in parallel

$$\bar{Y} = \bar{Y}_1 + \bar{Y}_2 + \dots \quad (\text{AI.26})$$

or

$$\frac{1}{\bar{Z}} = \frac{1}{\bar{Z}_1} + \frac{1}{\bar{Z}_2} + \dots \quad (\text{AI.27})$$

It is to be observed here that the impedance form (Eq. (AI.24)) is more convenient for series circuit and the admittance form (Eq. (AI.26)) for parallel circuits. Further, it must be reminded here that series/parallel combinations involve complex number computations.

AI.7 | POWER IN AC CIRCUIT (SINGLE-PHASE)

Consider Fig. AI.12 in which a single-phase source supplies a load of impedance $Z \angle \theta$. Then in instantaneous form

$$v = \sqrt{2} V \sin \omega t$$

$$i = \sqrt{2} I \sin(\omega t - \theta)$$

The instantaneous power fed to the load is given by

$$\begin{aligned} p = vi &= 2VI \sin \omega t \sin(\omega t - \theta) \\ &= VI[\cos \theta - \cos(2\omega t - \theta)] \end{aligned} \quad (\text{AI.28})$$

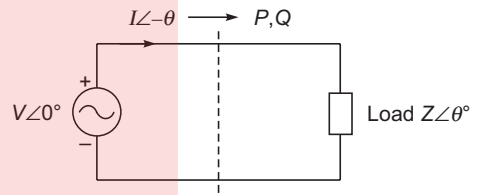


Fig. AI.12

It is easily seen that the instantaneous power has a constant component and an oscillating component of double frequency (2ω). The average power (*active*) being fed to the load over a complete cycle is

$$P = VI \cos \theta \text{ W} \quad (\text{AI.29})$$

Large power can be expressed in units of kilowatts or megawatts (kW/MW). We define

$$\cos \theta = \text{power factor (pf)} \quad (\text{AI.30})$$

and it could be lagging or leading depending upon whether current lags or leads the voltage.

Power can also be interpreted by means of the phasor diagram of Fig. AI.13(a) wherein the current component $I \cos \theta$ in phase with voltage supplies active power to load. The second component of current $I \sin \theta$ supplies *reactive* power to the load, i.e.

$$Q = VI \sin \theta \quad \text{VAR (VA reactive)} \quad (\text{AI.31})$$

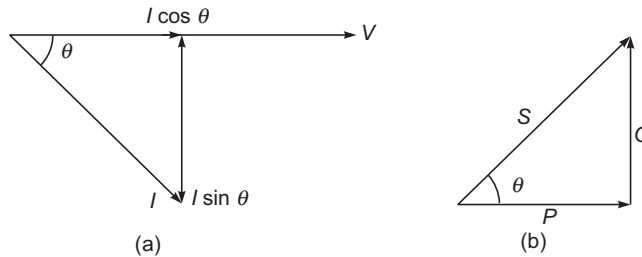


Fig. AI.13

which can also be measured in practical units of kVAR/MVAR (kilo volt-amp reactive/mega volt-amp reactive) for large power equipment. Reactive power will be taken as *positive* for *lagging* load and *negative* for *leading* load.

AI.8 | COMPLEX POWER

It is convenient to express power in complex form

$$\bar{S} = \bar{V} \bar{I}^* \quad (\text{AI.32})$$

Let $\bar{V} = V e^{j\phi}$; $\bar{I} = I e^{j(\phi-\theta)}$; $\bar{I}$ lags $\bar{V}$ by θ

$$\begin{aligned} \text{then } \bar{S} &= V e^{j\phi} \cdot I e^{-j(\phi-\theta)} \\ &= V I e^{j\theta} \\ &= V I \cos \theta + j V I \sin \theta \end{aligned} \quad (\text{AI.33})$$

$$= P + jQ \quad (\text{AI.34})$$

It may be seen that Q is positive for θ lagging.

The phasor diagram of a complex power corresponding to Eq. (AI.34) is drawn in Fig. AI.13(b) wherein the sign of θ is reversed with respect to that of Fig. AI.13(a) because of conjugating of current in Eq. (AI.32). It follows from the phasor diagram that

$$\begin{aligned} S &= \sqrt{P^2 + Q^2} \\ &= \text{VA delivered to load} \end{aligned} \quad (\text{AI.35})$$

$$\text{pf} = \cos \theta = \cos \tan^{-1} \frac{Q}{P}; \quad \begin{array}{l} \text{lagging for positive } Q, \\ \text{leading for negative } Q \end{array} \quad (\text{AI.36})$$

For the series RL circuit of Fig. AI.8

$$\begin{aligned} \bar{S} &= \bar{V} \bar{I}^* \\ &= \bar{Z} \bar{I} \bar{I}^* \\ &= (R + jX_L) I^2 = R I^2 + jX_L I^2 \end{aligned} \quad (\text{AI.37})$$

Hence

$$P = R I^2 = \text{real power consumed in resistive element}$$

$$Q = X_L I^2 = \text{reactive power consumed in reactive (inductive) element}$$

AI.9 | PARALLEL RLC CIRCUIT

Consider the parallel RLC circuit of Fig. AI.14(a). It immediately follows that

$$\bar{I} = GV - j(B_L - B_C)V \quad (\text{AI.38})$$

whose phasor diagram is depicted in Fig. AI.14(b). Here,

$$\theta = \tan^{-1} \left(\frac{B_L - B_C}{G} \right) \quad (\text{AI.39})$$

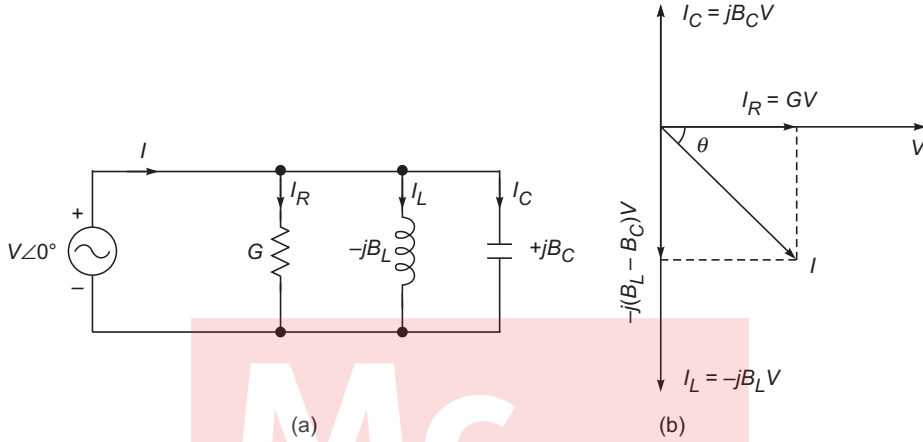


Fig. AI.14 Parallel RLC circuit

lagging for $B_L > B_C$ and leading for $B_L < B_C$. The power factor of the circuit is expressed as

$$\cos \theta = \cos \tan^{-1} \left(\frac{B_L - B_C}{G} \right) \quad (\text{AI.40})$$

It is easily seen that for a given load (G and B_L), the power factor can be adjusted by B_C of the capacitance.

The complex power consumed by the circuit is

$$\bar{S} = GV^2 + j(B_L - B_C)V^2 \quad (\text{AI.41})$$

Thus with the real power fixed (GV^2), the reactive power can be controlled by adjusting the capacitance (B_C) which, in other words, means that the circuit power factor can be controlled.

Three-Phase Systems

Three-phase* devices and systems are almost universally adopted for large and medium generation, transmission and utilization of electric energy. It is a well-proven fact that 3-phase systems are more economical and have definite operational advantages compared to single-phase systems.

AII.1 | THREE-PHASE VOLTAGES AND CURRENTS

A set of three voltages/emfs which are equal in magnitude and have a relative progressive phase difference[†] of $2\pi/3 (=120^\circ)$ constitute *balanced* 3-phase voltages/emfs. The 3-phase voltages are otherwise said to be *unbalanced*. The same definition applies to a set of three currents. Figure AII.1 shows three windings (*phases*) whose voltages/emfs form a balanced set. In instantaneous form these can then be expressed as

$$e_{aa'} = \sqrt{2} E \sin \omega t \quad (\text{AII.1a})$$

$$e_{bb'} = \sqrt{2} E \sin (\omega t - 120^\circ) \quad (\text{AII.1b})$$

$$e_{cc'} = \sqrt{2} E \sin (\omega t - 240^\circ) \quad (\text{AII.1c})$$

The waveforms of these voltages are drawn in Fig. AII.2. In the case being considered the voltage of phase *a* leads that of *b* and the voltage of phase *b* leads that of *c*; a fact which is remembered by saying that the 3-phase voltages have a *phase sequence abc*. The phase sequence could alternatively be *acb*, i.e. *a* lags *b* lags *c*.

In phasor form Eqs (AII.1a, b, c) can be written as

$$\bar{E}_{aa'} = E \angle 0^\circ \quad (\text{AII.2a})$$

$$\bar{E}_{bb'} = E \angle -120^\circ \quad (\text{AII.2b})$$

$$\bar{E}_{cc'} = E \angle -240^\circ \quad (\text{AII.2c})$$

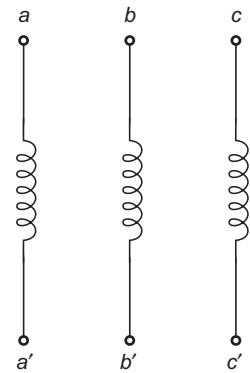


Fig. AII.1

* For the last decade interest is also being expressed in 6-phase systems.

[†] A balanced set of *q*-phase voltages have a relative phase difference of

$2\pi/q$ ($q \neq 2$) and $\pi/2$ for $q = 2$.

Consider now the instantaneous voltage sum

$$e_{aa'} + e_{bb'} + e_{cc'} = \sqrt{2} E [\sin \omega t + \sin (\omega t - 120^\circ) + \sin (\omega t - 240^\circ)]$$

$$\text{or} \quad e_{aa'} + e_{bb'} + e_{cc'} = 0 \quad (\text{AII.3})$$

It is therefore seen that the sum of balanced 3-phase voltages (or currents) is zero at all time. This is also evident from the waveforms of Fig. AII.2. In phasor form this implies

$$\bar{E}_{aa'} + \bar{E}_{bb'} + \bar{E}_{cc'} = 0 \quad (\text{AII.4})$$

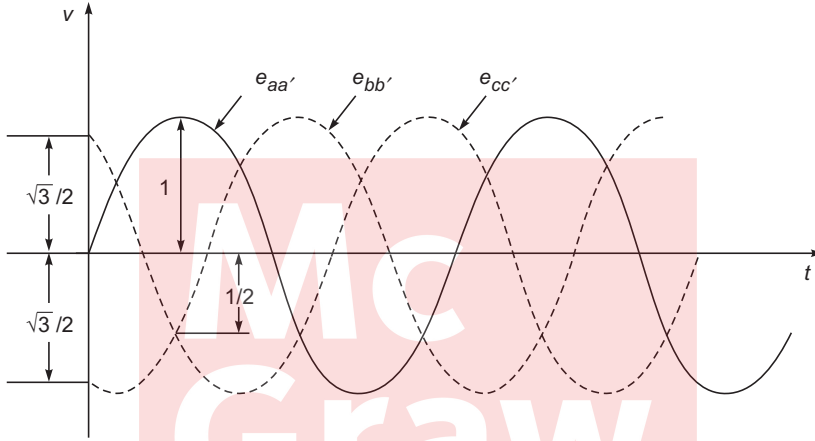


Fig. AII.2

The 3-phase windings of Fig. AII.1 can be connected in two ways illustrated below.

AII.2 | STAR CONNECTION

In this connection the three similar polarity ends (say a' , b' , c') of the phasor windings are joined together to form the *neutral point* while three lines are taken from the other ends (a , b , c) as shown in Fig. AII.3. One can now speak in terms of phase voltages and currents in phases (windings) and line voltages and currents and their interrelationships.

It is evident from Fig. AII.3 that

$$\bar{E}_{an} = \bar{E}_a = \bar{E}_{aa'} = E \angle 0^\circ \quad (\text{AII.5a})$$

$$\bar{E}_{bn} = \bar{E}_b = \bar{E}_{bb'} = E \angle -120^\circ \quad (\text{AII.5b})$$

$$\bar{E}_{cn} = \bar{E}_c = \bar{E}_{cc'} = E \angle -240^\circ \quad (\text{AII.5c})$$

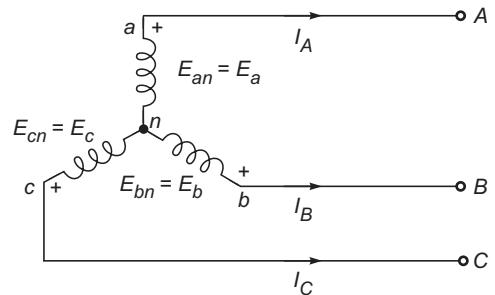


Fig. AII.3 Star connection

These phase voltages are indicated in the phasor diagram of Fig. AII.4. From the connection diagram of Fig. AII.3 it is found that the voltage between a and b (or A and B) is

$$\bar{E}_{ab} = \bar{E}_{an} - \bar{E}_{bn}$$

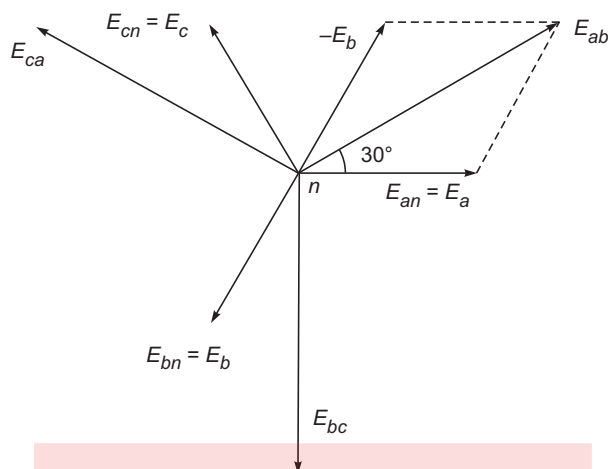


Fig. AII.4

It immediately follows from the phasor operation of Fig. AII.4 that

$$\bar{E}_{ab} = \sqrt{3} \bar{E}_{an} \angle 30^\circ = \sqrt{3} E \angle 30^\circ \quad (\text{AII.6a})$$

Similarly,

$$\begin{aligned} \bar{E}_{bc} &= \bar{E}_{bn} - \bar{E}_{cn} \\ &= \sqrt{3} \bar{E}_{bn} \angle 30^\circ = \sqrt{3} E \angle -90^\circ \end{aligned} \quad (\text{AII.6b})$$

$$\begin{aligned} \bar{E}_{ca} &= \bar{E}_{cn} - \bar{E}_{an} \\ &= \sqrt{3} \bar{E}_{cn} \angle 30^\circ = \sqrt{3} E \angle 150^\circ \end{aligned} \quad (\text{AII.6c})$$

It is seen from the complete phasor diagram of Fig. AII.4 that for balanced phase voltages, line voltages also form a balanced set advanced in phase by 30° from the phase voltages. In terms of magnitudes

$$E(\text{line-to-line}) = \sqrt{3} E(\text{phase})$$

or

$$E_L = \sqrt{3} E_P \quad (\text{AII.7})$$

It is easily seen from Fig. AII.3 that

$$\bar{I}_A = \bar{I}_a \quad (\text{AII.8a})$$

$$\bar{I}_B = \bar{I}_b \quad (\text{AII.8b})$$

$$\bar{I}_C = \bar{I}_c \quad (\text{AII.8c})$$

Thus if the phase currents are a balanced set, the line currents also constitute a balanced set as shown in Fig. AII.5. In a balanced system, θ is the phase angle difference between the voltages and currents of corresponding phases. In terms of magnitudes

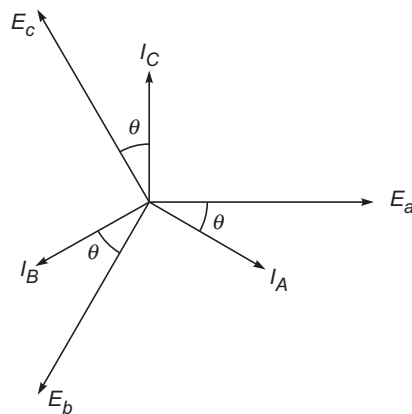


Fig. AII.5

$$I(\text{line}) = I(\text{phase})$$

or

$$I_L = I_P \quad (\text{AII.9})$$

Being a balanced set

$$\bar{I}_A + \bar{I}_B + \bar{I}_C = 0 \quad (\text{AII.10})$$

However, with isolated neutral in a star connection, even when currents are unbalanced, Eq. (AII.10) must hold.

To summarize, the magnitude relations in a star connection:

$$E_L = \sqrt{3} E_P; \quad I_L = I_P \quad (\text{AII.11})$$

AII.3 | DELTA CONNECTION

The phase windings of Fig. AII.1 can be connected[‡] in delta by joining ac' , ba' and cb' and taking out three lines from the nodes so formed. It is immediately seen that the voltage round the closed loop formed by the delta connection is

$\bar{E}_{aa'} + \bar{E}_{bb'} + \bar{E}_{cc'} = 0$ (being a balanced set of phase voltages and hence delta does not constitute a short circuit).

In delta connection, phase and line currents are related as shift (Fig. AII.6)

$$\bar{I}_A = \bar{I}_{ba} - \bar{I}_{ac} = \sqrt{3} \bar{I}_{ba} \angle -30^\circ \quad (\text{AII.12a})$$

$$\bar{I}_B = \bar{I}_{cb} - \bar{I}_{ba} = \sqrt{3} \bar{I}_{cb} \angle -30^\circ \quad (\text{AII.12b})$$

$$\bar{I}_C = \bar{I}_{ac} - \bar{I}_{cb} = \sqrt{3} \bar{I}_{ac} \angle -30^\circ \quad (\text{AII.12c})$$

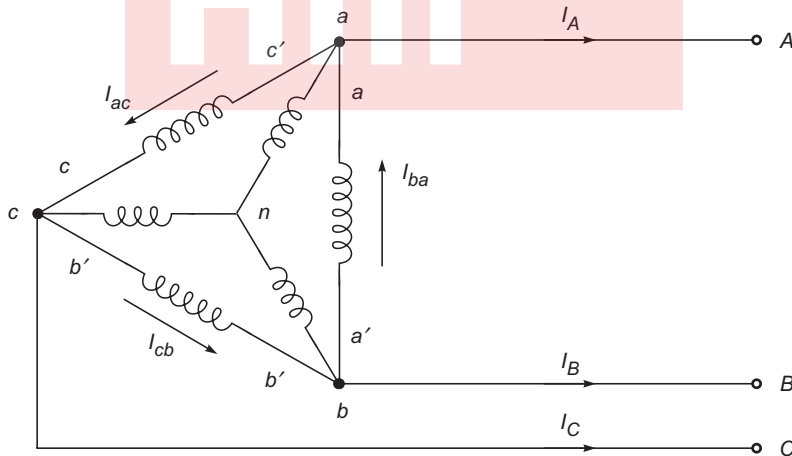


Fig. AII.6 Delta connection

[‡]Other connections for delta are possible (see Sec. 3.13).

These relationships are also shown in the phasor diagram of Fig. AII.7. Obviously when phase currents are balanced, so are line currents and vice versa. In terms of magnitudes

$$I_L = \sqrt{3} I_P \quad (\text{AII.13})$$

It is easily concluded that

$$E_L = E_P \quad (\text{AII.14})$$

The reader should compare these magnitude relationships (Eqs (AII.13) and (AII.14)) with those of star connection (Eq. (AII.11)).

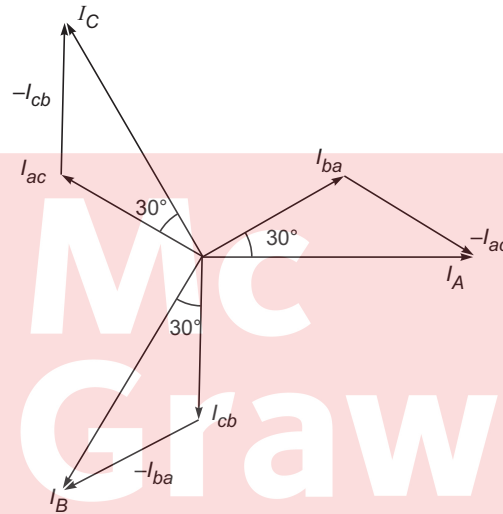


Fig. AII.7

AII.4 | EQUIVALENT STAR OF A DELTA CONNECTION

It is usually convenient to work in terms of equivalent star of a delta connection. Such equivalence is shown dotted in Fig. AII.6. The voltage and current magnitude relationships already given would apply. The phase-angle relationship between delta and equivalent star voltages are brought out by Fig. AII.8. The corresponding phase relationship between currents follows from Fig. AII.7.

Equivalent delta of a star connection also easily follows from Figs AII.6 and AII.8.

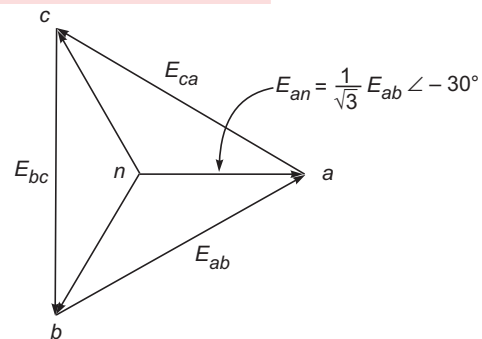


Fig. AII.8

AII.5 | POWER

Phase voltages and currents in a balanced 3-phase circuit (star or delta) can be written in the instantaneous form as

$$e_a = \sqrt{2} E_P \sin \omega t \quad (\text{AII.15a})$$

$$e_b = \sqrt{2} E_P \sin (\omega t - 120^\circ) \quad (\text{AII.15b})$$

$$e_c = \sqrt{2} E_P \sin (\omega t - 240^\circ) \quad (\text{AII.15c})$$

and

$$i_a = \sqrt{2} I_P \sin (\omega t - \theta) \quad (\text{AII.16a})$$

$$i_b = \sqrt{2} I_P \sin (\omega t - \theta - 120^\circ) \quad (\text{AII.16b})$$

$$i_c = \sqrt{2} I_P \sin (\omega t - \theta - 240^\circ) \quad (\text{AII.16c})$$

where θ = phase angle between phase current and phase voltage.

The phasor relationship between balanced phase voltages and currents is shown in Fig. AII.5.

The instantaneous power in each phase is

$$p_a = e_a i_a = E_P I_P [\cos \theta - \cos (2\omega t - \theta)] \quad (\text{AII.17a})$$

$$p_b = e_b i_b = E_P I_P [\cos \theta - \cos (2\omega t - \theta - 240^\circ)] \quad (\text{AII.17b})$$

$$p_c = e_c i_c = E_P I_P [\cos \theta - \cos (2\omega t - \theta - 480^\circ)] \quad (\text{AII.17c})$$

The total instantaneous power is

$$p = p_a + p_b + p_c = 3 E_P I_P \cos \theta \quad (\text{AII.18})$$

Notice that the sum of the three cosine terms, which have a progressive phase difference of 120° , is zero. From Eq. (AII.18) it is found that the total instantaneous power in a balanced 3-phase system is constant and is equal to three times the average power per phase. This observation is in contrast to power in a single-phase system (Eq. (AI.28)) which has a double-frequency oscillating component. Constancy of instantaneous power in a balanced 3-phase system affords the important advantage of uniform torque in rotating machines with the absence of pulsating torque which would be present in single-phase machines. This is one of the many but important reasons in universal adoption of 3-phase system except in low power applications.

Thus in star/delta balanced system

$$P = 3 E_P I_P \cos \theta \quad (\text{AII.19})$$

Converting phase values to line values

$$P = \sqrt{3} E_L I_L \cos \theta = 3 I_P^2 R_p \quad (\text{AII.20})$$

Also

$$Q = \sqrt{3} E_L I_L \sin \theta = 3 I_P^2 X_p \quad (\text{AII.21})$$

where R_p = equivalent series resistance of load and X_p = equivalent series reactance of the load.

The phase angle of the load is given by

$$\theta = \tan^{-1} \frac{X_P}{R_P} = \cos^{-1} \frac{R_P}{Z_P} = \tan^{-1} \frac{Q}{P} \quad (\text{AII.22})$$

In a balanced 3-phase system each phase has the same power factor.

AII.6 THREE-PHASE CIRCUITS

Figure AII.9 shows a typical balanced 3-phase system—source, transmission line and load. The source and load are assumed to be star-connected though either or both of them could be delta-connected. The two neutrals are isolated; even if these are connected (shown by dotted line) the current in the neutral connection will be zero by virtue of the balanced currents ($\bar{I}_n = \bar{I}_a + \bar{I}_b + \bar{I}_c = 0$). For the reference phase a (loop shown)

$$\bar{E}_a = (\bar{Z}_{TL} + \bar{Z}_L) \bar{I}_a \quad (\text{AII.23})$$

Because of zero current in the neutral connection, if present, its impedance would not appear in Eq. (AII.23). The currents and voltages in other phases would have the same magnitude as given by Eq. (AII.23) but would have a progressive shift in phase of 120° . Equation (AII.23) corresponds to the single-phase circuit shown in Fig. AII.10 whose solution completely determines the solution of the 3-phase circuit of Fig. AII.9. If the load or any part of the system were unbalanced, one must proceed on a 3-phase basis by writing three loop equations.

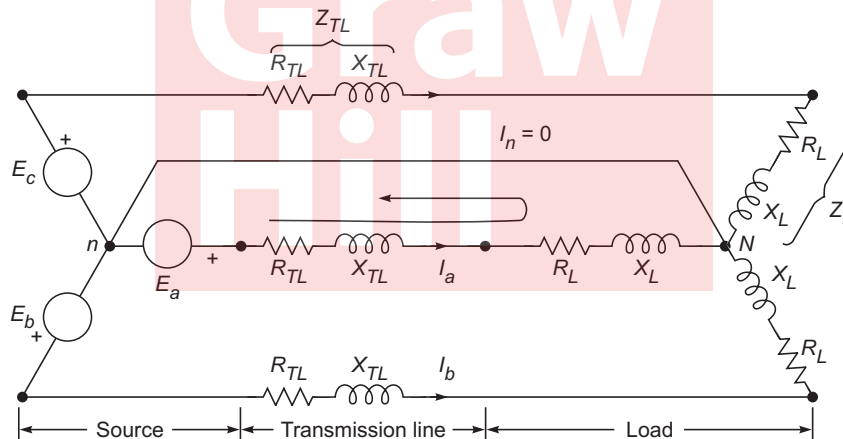


Fig. AII.9 Balanced 3-phase system

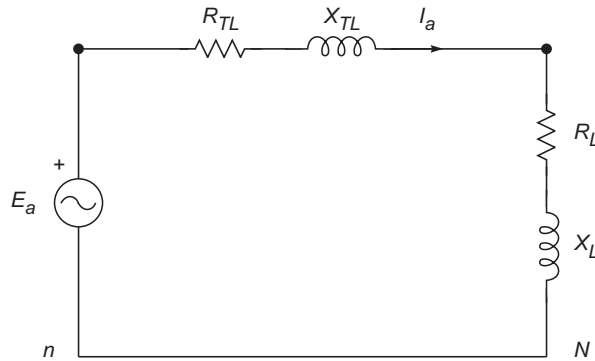


Fig. AII.10

Analysis of a balanced 3-phase system is thus extremely simple and proceeds on a single-phase basis. Where deltas are involved, one can either proceed directly or convert these to equivalent star. This procedure will be illustrated in Ex. AII.1.

AII.7 STAR-DELTA CONVERSION

Figure AII.11 shows three delta-connected impedances whose equivalent star, shown dotted, is required to be found, by equating the impedances between all three 2-terminal pairs by turn. Between terminals a and b .

$$\begin{aligned}\bar{Z}_a + \bar{Z}_b &= \bar{Z}_{ab} || (\bar{Z}_{bc} + \bar{Z}_{ca}) \\ &= \frac{\bar{Z}_{ab} (\bar{Z}_{bc} + \bar{Z}_{ca})}{\bar{Z}_{ab} + \bar{Z}_{bc} + \bar{Z}_{ca}}\end{aligned}\quad (\text{AII.24})$$

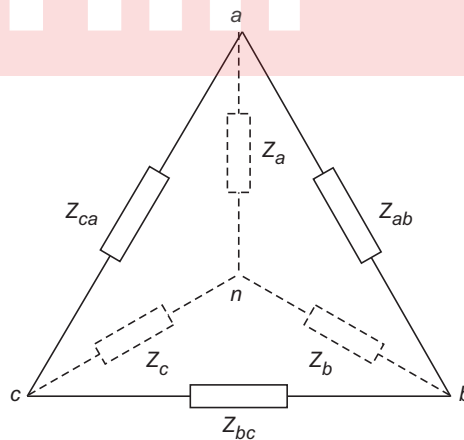


Fig. AII.11

Similarly

$$\bar{Z}_b + \bar{Z}_c = \frac{\bar{Z}_{bc}(\bar{Z}_{ca} + \bar{Z}_{ab})}{\bar{Z}_{ab} + \bar{Z}_{bc} + \bar{Z}_{ca}} \quad (\text{AII.25})$$

$$\bar{Z}_c + \bar{Z}_a = \frac{\bar{Z}_{ca}(\bar{Z}_{ab} + \bar{Z}_{bc})}{\bar{Z}_{ab} + \bar{Z}_{bc} + \bar{Z}_{ca}} \quad (\text{AII.26})$$

Solving

$$\bar{Z}_a = \frac{\bar{Z}_{ab} \bar{Z}_{ca}}{\bar{Z}_{ab} + \bar{Z}_{bc} + \bar{Z}_{ca}} \quad (\text{AII.27})$$

$$\bar{Z}_b = \frac{\bar{Z}_{ab} \bar{Z}_{bc}}{\bar{Z}_{ab} + \bar{Z}_{bc} + \bar{Z}_{ca}} \quad (\text{AII.28})$$

$$\bar{Z}_c = \frac{\bar{Z}_{ca} \bar{Z}_{bc}}{\bar{Z}_{ab} + \bar{Z}_{bc} + \bar{Z}_{ca}} \quad (\text{AII.29})$$

Solving Eqs (AII.24)–(AII.26), equivalent delta impedance of a star can be obtained. The results are reproduced below:

$$\bar{Z}_{ab} = \frac{\bar{Z}_a \bar{Z}_b + \bar{Z}_b \bar{Z}_c + \bar{Z}_c \bar{Z}_a}{\bar{Z}_c} \quad (\text{AII.30})$$

$$\bar{Z}_{bc} = \frac{\bar{Z}_a \bar{Z}_b + \bar{Z}_b \bar{Z}_c + \bar{Z}_c \bar{Z}_a}{\bar{Z}_a} \quad (\text{AII.31})$$

$$\bar{Z}_{ca} = \frac{\bar{Z}_a \bar{Z}_b + \bar{Z}_b \bar{Z}_c + \bar{Z}_c \bar{Z}_a}{\bar{Z}_b} \quad (\text{AII.32})$$

If the impedances are *balanced*, it follows from Eqs (AII.27)–(AII.29) or Eqs (AII.30)–(AII.32) that

$$\bar{Z}_\Delta = 3 \bar{Z}_Y \quad (\text{AII.33})$$

Example AII.1 A balanced delta-connected load of impedance $(4 + j3) \Omega/\text{phase}$ is connected to a 3-phase 230 V supply. Find the line current, power factor, power, reactive VA and total VA.

Solution

With reference to Fig. AII.2

$$V_P = V_L = 230 \text{ V}$$

$$I_P = \frac{230 \angle 0^\circ}{4 + j3} = 46 \angle -36.9^\circ \text{ A}$$

$$pf = \cos 36.9^\circ = 0.8 \text{ lagging}$$

$$I_L = \sqrt{3} I_P = \sqrt{3} \times 46 = 79.6 \text{ A}$$

$$P = \sqrt{3} V_L I_L \cos \theta$$

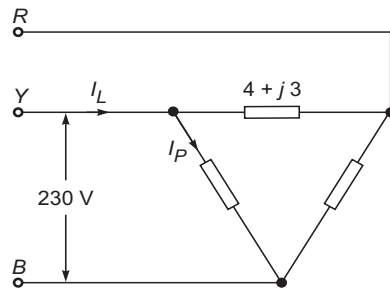


Fig. AII.12

$$= \sqrt{3} \times 230 \times 79.6 \times 0.8 = 25.36 \text{ kW}$$

$$Q = \sqrt{3} V_L I_L \sin \theta = \sqrt{3} \times 230 \times 79.6 \times 0.6 = +19.02 \text{ kVAR}$$

$$S = \sqrt{P^2 + Q^2} = 31.7 \text{ kVA}$$

Solution by equivalent star

With reference to Fig. AII.13

$$\bar{Z}_Y = \frac{1}{3} (4 + j3)$$

$$V_P = 230 / \sqrt{3} = 132.8 \text{ V}$$

$$\bar{I}_L = \bar{I}_P = \frac{132.8 \angle 0^\circ}{\frac{1}{3} (4 + j3)} = 79.6 \angle -36.9^\circ$$

All other results now follow.

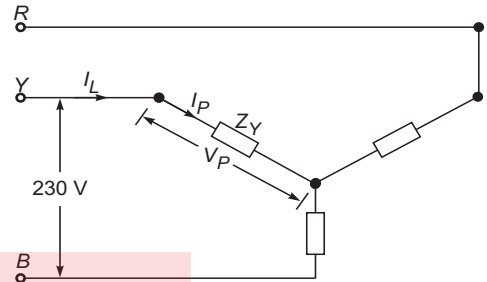


Fig. AII.13

Example AII.2 A 3-phase, 400 V source (terminal voltage assumed constant independent of load) supplies a load with an equivalent-star impedance of $(40 + j12) \Omega$ /phase through a transmission line impedance $(0.2 + j0.8) \Omega$ /line. Compute (a) the line current; (b) the load voltage; (c) the power, reactive power and VA consumed by the load; (d) the power and reactive power loss in the line; and (e) the power, reactive power and VA supplied by the source.

Solution

Figure AII.14 is the per phase circuit of the system (star basis)

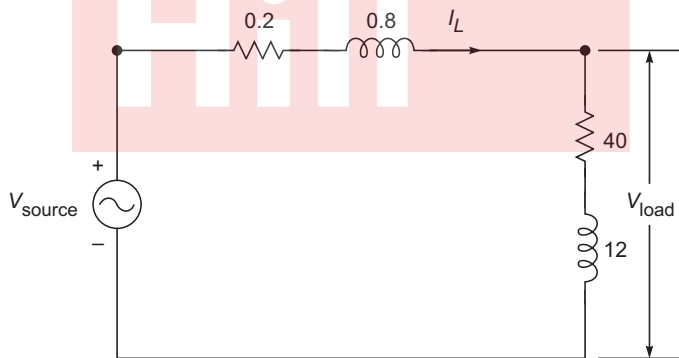


Fig. AII.14

$$V_{\text{source}} = 400 / \sqrt{3} = 231 \text{ V}$$

$$\bar{Z} \text{ (total)} = (0.2 + j0.8) + (40 + j12) = 40.2 + j12.8 = 42.2 \angle 17.7^\circ$$

$$I_L = \frac{231}{42.2} = 5.47 \text{ A}$$

$$\begin{aligned}\bar{Z}_{\text{load}} &= 40 + j12 = 41.8 \angle 16.7^\circ \\ V_{\text{load}} &= 5.47 \times 41.8 = 228.6 \text{ V} \quad (\text{phase-to-neutral}) \\ &= 396 \text{ V} \quad (\text{line-to-line}) \\ P_{\text{load}} &= 3 \times (5.47)^2 \times 40 = 3.59 \text{ kW} \\ Q_{\text{load}} &= 3 \times (5.47)^2 \times 12 = 1.077 \text{ kVAR} \\ S_{\text{load}} &= \sqrt{(3590)^2 + (1077)^2} = 3.748 \text{ kVA} \\ P_{\text{line}} &= 3 \times (5.47)^2 \times 0.2 = 18 \text{ W} \\ Q_{\text{line}} &= 3 \times (5.47) \times 0.8 = 72 \text{ VAR} \\ P_{\text{source}} &= 3.59 + 0.018 = 3.608 \text{ kW} \\ Q_{\text{source}} &= 1.077 + 0.072 = 1.149 \text{ kVAR} \\ S_{\text{source}} &= \sqrt{(3.608)^2 + (1.149)^2} = 3.786 \text{ kVA}\end{aligned}$$

PRACTICE PROBLEMS

- AI1.1** A balanced star-connected load is supplied from a symmetrical 3-phase, 400 V system. The current in each phase is 30 A and lags 30° behind the phase voltage. Find (a) the phase voltage, and (b) power and reactive VA drawn by the load. Draw the phasor diagram.
- AI1.2** A balanced delta-connected load of $15 + j12 \Omega$ /phase is connected to a 3-phase, 230 V supply. Find the line current, power factor, power, reactive VA and total VA.
- AI1.3** A symmetrical 3-phase, 400 V system supplies a balanced delta-connected load. The current in each delta branch is 25 A and its phase angle is 30° lagging (with respect to voltage across the branch). Find the line current and the total power. Draw the phasor diagram showing currents and voltages.
- AI1.4** A balanced 3-phase, star-connected load of 20 kW takes a leading current of 50 A with a line voltage of 400 V, 50 Hz. Find the per phase circuit parameters of the load.
- AI1.5** Three-star-connected impedances of $\bar{Z}_1 = 20 + j40 \Omega$ per phase are in parallel with three delta-connected impedances of $\bar{Z}_2 = 30 \Omega - j21 \Omega$ per phase across a 3-phase supply of 400 V. Find the line current, power factor, power and reactive VA of the combination.
- AI1.6** Figure AP11.6 shows what is known as a *one-line diagram* of a power system. Impedances indicated on the lines are per line (phase) values. The load requires 500 kW at 0.8 lagging power factor. Generator G_1 has a terminal voltage of 10 kV and supplies (output) 250 kW at 0.75 power factor lagging.

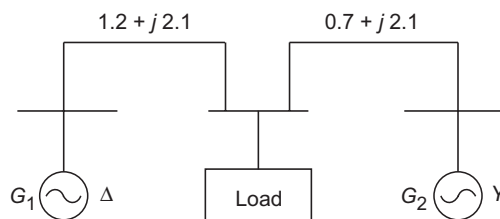


Fig. AP11.6

Find the load voltage and terminal voltage and power and reactive VA supplied by G_2 .

Special Topics in Transformers

AIII.1 | FREQUENCY RESPONSE

In electronic amplifiers, transformers are used at the output stage with frequency being variable over a band. In such variable-frequency applications, the frequency response of a transformer is of great significance. Frequency response is the V_2/V_1 behaviour for fixed V_1 and load but variable frequency. A flat frequency response over the frequency band of interest is most desirable. The corresponding variation of the phase angle of the load voltage (V_2) with respect to the source voltage (V_1) is called the phase response. A small phase angle is acceptable.

The complete frequency range can be divided into three parts—low frequency (LF) range, intermediate frequency (IF) range and high frequency (HF) range. The transformer is so designed that the equivalent circuit of Fig. 3.14(c) is applicable over the IF range. Here the magnetizing reactance of the transformer is large and does not affect its voltage-ratio behaviour and the series reactances of the windings are small enough so that V_2/V_1 remains substantially constant as shown in Fig. AIII.1. In the LF range the transformer draws large magnetizing current causing heavy voltage drop in the primary winding such that V_2/V_1 reduces continuously with decreasing frequency. The output voltage V_2 is of course zero for dc ($f = 0$). In the HF range the effect of inter-turn, turn-to-ground and stray capacitances can no longer be neglected.

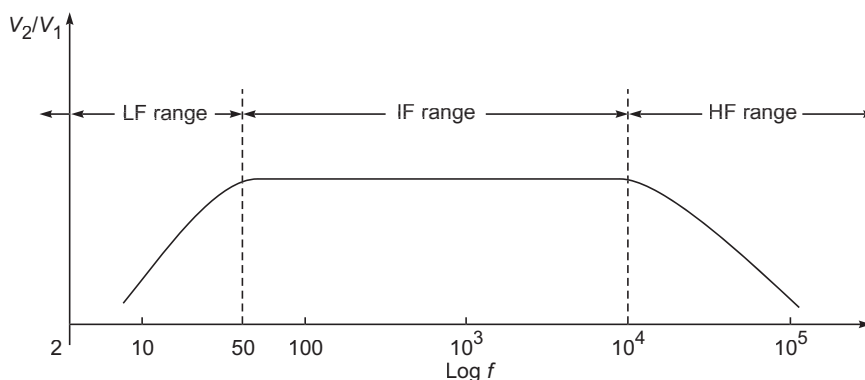


Fig. AIII.1 Frequency response (V_2/V_1 versus $\log f$) of a transformer

The combined effects of increasing series inductive reactance and decreasing shunt capacitive reactance with increasing frequency cause the output voltage to drop sharply. The complete frequency response of a transformer on a logarithmic frequency scale is illustrated in Fig. AIII.1.

AIII.2 | PULSE TRANSFORMERS

Transformers used in pulse or digital circuits are known as pulse transformers. These normally operate from a discontinuous source of excitation because of which the distributed capacitance of the transformer winding must be taken into account.

For a pulse transformer, the average power is normally very low, but the peak power is quite large. However, recently, medium- and low-peak power pulse transformers have been developed for many electronic and control applications, such as in the gate firing circuits of SCRs. It is important that the transformer reproduces the input pulse (normally rectangular) as accurately as possible at its secondary terminals. Figure AIII.2(a) depicts a square wave input pulse. The pulse width varies from a fraction of a microsecond to about 25 μs and quite sometime will elapse before the next pulse appears. To find the output waveform, a transient analysis needs to be resorted to. Figure AIII.2(b) shows the corresponding output pulse wherein various pulse transformer parameters are defined.

The analysis is carried out normally by solving the differential equations in a piecewise manner using the equivalent circuit of a transformer including stray capacitance. Owing to the stray capacitance, there will usually be damped oscillations. Leakage inductance is kept to a minimum value if the shortest rise time is desired.

Pulse transformers are quite small in size and have comparatively few turns so that the leakage inductance is minimized. The cores are constructed of ferrites or wound strips of high permeability alloys such as permalloy. This helps in achieving a high magnetizing inductance so that the tilt of the pulse top is kept within permissible limits. Since the time interval between pulses is long as compared to the pulse duration, the pulse transformer can carry high pulse-power levels.

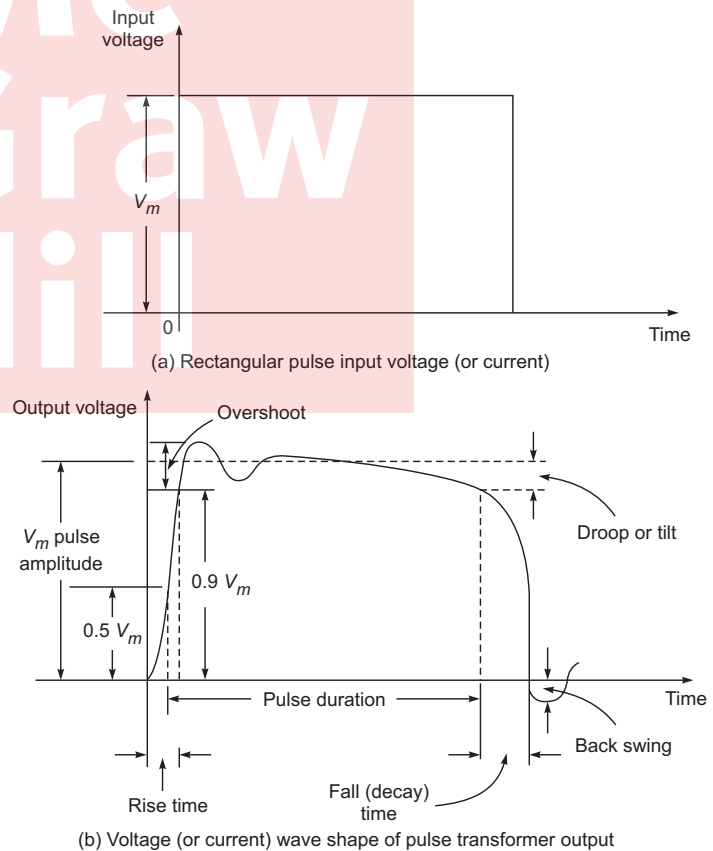


Fig. AIII.2

APPENDIX

IV

Cross-Field Machines

At the output stage of a control system, a low power electrical control signal is required to control large electric power necessitating large power amplification. Presently, SCR power amplifiers serve this purpose excellently. Prior to the advent of SCRs, large power amplification with low associated time constants were devised by means of a family of dc machines called cross-field machines. Though an ordinary dc generator is a power amplifier—a small field power controls large power at armature terminals—their power amplification is not sufficiently high and their response is sluggish. Cross-field machines which overcome both these drawbacks of dc generators have windings on d - and q -axes of the stator and brush pairs on the rotor forming both q - and d -axes mmf directions fully utilizing the potentiality of the dc machine. Cross-field machines have two brush sets per pair of poles. Thereby they use the armature mmf to provide most of the excitation and achieve high power gains. In fact, cross-field machines behave like two dc generators in cascade accommodated in a single machine frame. There are several types of cross-field generators—metadyne, amplidyne and regulex. Here metadyne will be studied first and then the most commonly used form, the amplidyne, will be dealt with.

An ordinary dc generator is modified into a metadyne by providing an additional pair of brushes in the d -axis as shown in the schematic diagram of Fig. AIV.1. The q -axis brushes are shorted, while the output is obtained from the d -axis brushes. Control field winding mmf, F_f induces the q -axis emf which causes the short-circuit q -axis current I_q to flow. The q -axis mmf F_q thus established induces the d -axis emf causing the d -axis current I_d to flow through the load. The d -axis reaction mmf F_d , by Lenz's law, opposes the cause F_f . For a given F_f and load resistance, the system settles down to certain steady values of I_q and I_d . Any increase in load (decrease of load resistance) increases F_d which neutralizes more of F_f effectively reducing d -axis induced emf preventing the load current from increasing. Thus for a given F_f (control excitation), the output current of the metadyne remains substantially constant over a wide range of load variation, i.e. metadyne behaves as a constant current generator.

To convert a constant-current metadyne to *constant-voltage amplidyne*, a more useful machine, the d -axis mmf must be cancelled by means of a *compensating winding* located in the d -axis on the stator as shown in the schematic diagram of Fig. AIV.2. The d -axis output current flowing through the compensating winding develops an mmf, F_c , in opposition to that of the d -axis armature mmf. The negative feedback effect of the load current is therefore minimized. The d -axis flux is now mainly dependent on the field winding current, giving it wider control over the net d -axis flux. Because of the difference in the manner of space distribution of F_d and F_c , exact neutralization of F_d is never possible.

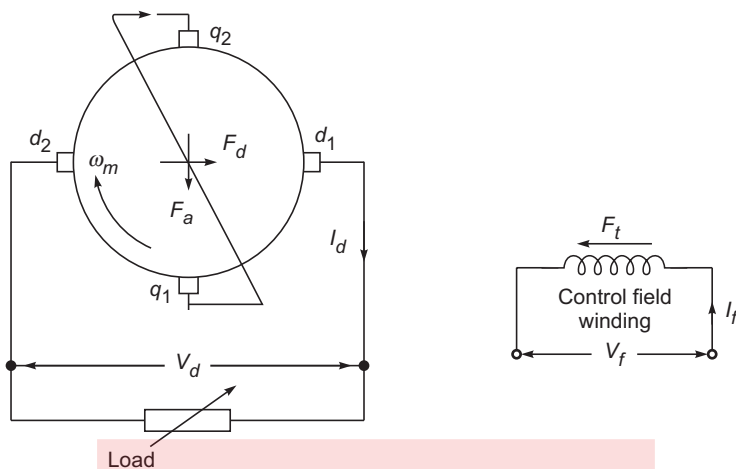


Fig. AIV.1 Schematic diagram of a metadyne generator

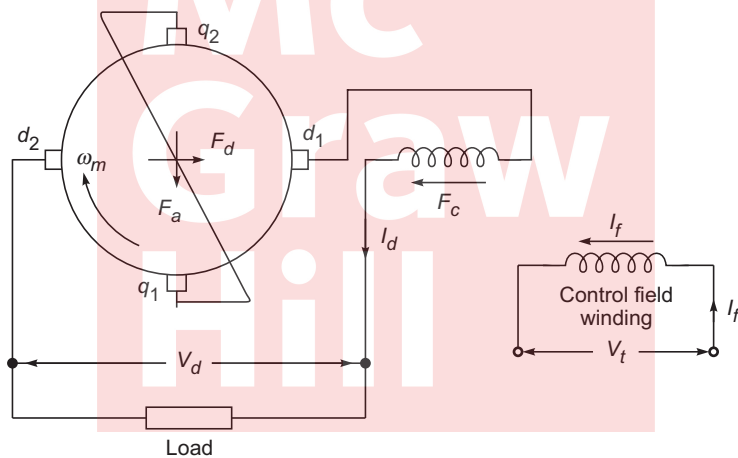


Fig. AIV.2 Schematic diagram of an amplidyne

By definition

$$C = \frac{\text{effective compensation turns}}{\text{effective armature turns}}$$

The effect of compensating factor C on the load characteristic of machine is depicted in Fig. AIV.3. For metadyne, $C = 0$ and for an amplidyne, $C = 1$, giving an amplidyne a characteristic like that of a separately excited generator. It is clear from Fig. AIV.3 that a metadyne should never be operated on open-circuit and an amplidyne on short-circuit.

To create a given F_q with smaller I_q , a series connected q -axis winding is usually provided on the stator of an amplidyne. The d - and q -axes currents flowing on the armature of an amplidyne tend to cancel each other

considerably increasing the amplidyne rating over that of a dc generator for the same frame size. This fact also results in large power amplification ($V_d I_d / V_f I_f$) in the range of 10^5 .

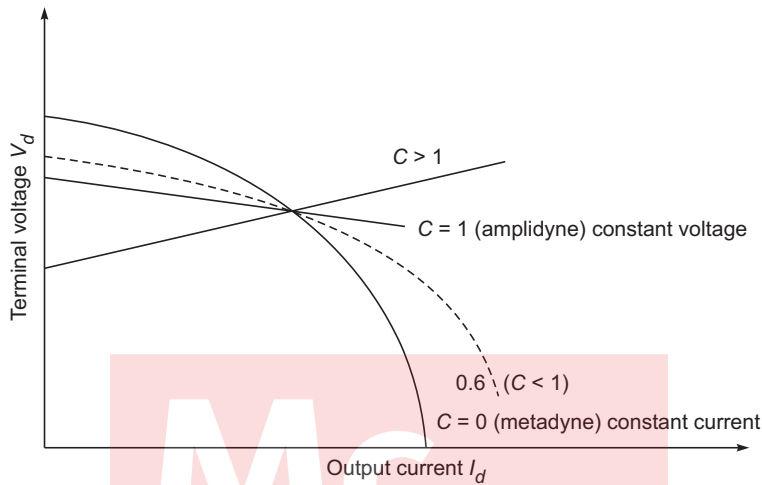


Fig. AIV.3 External characteristics of a cross-field generator

Amplidynes are used to provide controlled power amplification in a variety of control systems, e.g. as the voltage regulator in the excitation systems of large ac generators to introduce a buck or boost voltage in series with the field winding of the main exciter. An amplidyne may be used as a generator in the Ward-Leonard speed or position control system. In modern control systems these functions of an amplidyne have been taken over by SCR circuitry.

APPENDIX

V

ac Commutator Machines

It has been seen earlier that an induction motor has substantially constant speed in the operating range. In its cheap and rugged squirrel-cage form it also suffers from the fact of low starting torque. In Ch. 9 several methods of accomplishing speed control and obtaining good starting torque were studied. Another answer to these problems of the induction machine is the ac commutator machine. Because of the inherent problems of ac commutation, ac commutator motors have been manufactured in medium sizes only. With the advent of thyristors, the frequency-converting action of the commutator has now been taken over by SCR circuitry. Though ac commutator machines can still be found in older installations, they have for all practical purposes become a part of the history of electrical machines.

The underlying principles and basic constructional features of two once commonly used types of ac commutator machines—the Repulsion motor (single-phase) and Schrage motor (three-phase)—are described below.

AV.1 | REPULSION MOTORS

The repulsion motor is like a dc series motor with the rotor energized inductively. The stator slots house a single-phase exciting winding similar to the main winding of a single-phase induction motor. The rotor carries an ordinary distributed dc winding connected to the commutator at one end. The brushes are short-circuited to provide a closed current path as shown in Fig. AV.1.

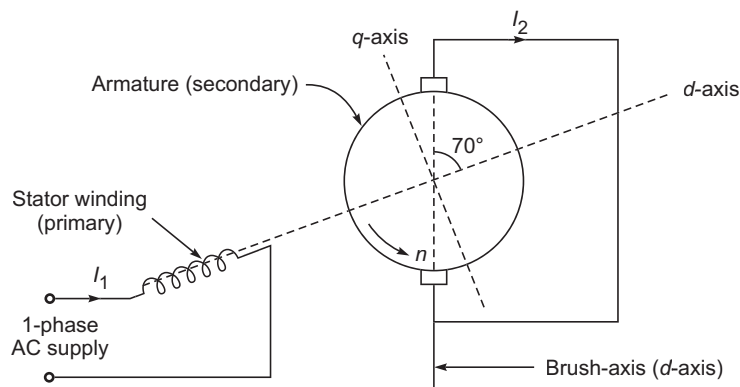


Fig. AV.1 Schematic diagram of a repulsion motor

If the brushes are located in the d -axis, the transformer emf in the effective armature coil will be maximum and so would be the short-circuit current I_2 ; but no torque will be produced as stator and rotor fields are aligned. If the brushes are placed in the q -axis, the transformer emf induced in the armature would add up to zero and so current $I_2 = 0$ and hence no torque is produced. As it is essential that both an armature current should flow and an angular displacement must exist between the two fields, the brushes are placed in an intermediate position making a large angle of about 70° with the d -axis. In a futile attempt to reduce copper losses, the rotor runs in a direction to orient itself with the q -axis, i.e. the direction of rotation is counter-clockwise for brushes displaced from the d -axis as shown in Fig. AV.1. This appears like repulsion between the two fields and hence the name. The direction of rotation will reverse if the brushes are displaced on the other side of the d -axis.

The unmodified repulsion motor has speed-torque characteristic similar to that of an ac series motor. It has a large starting torque and a tendency to overspeed at light load.

Repulsion-Induction Motor

The repulsion principle is often used merely to provide a high starting torque at a relatively low starting current. When the machine has accelerated close to the operating speed, it may be converted to a single-phase induction motor to provide essentially constant speed running characteristics. This is achieved if, in addition to an outer commutator winding, the rotor carries an inner squirrel-cage winding. The torques developed then are additive. Such a motor is called a *repulsion-induction motor*. At starting, the cage current is limited by its relatively high reactance and the commutator winding develops a major part of the torque, but during running conditions the cage torque increases, compensating for the natural drop in the speed-torque characteristic of the commutator winding.

Repulsion-Start Induction-Run Motor

Constancy of speed and high starting torque can be achieved in a repulsion motor by providing a centrifugal force-operated device to short-circuit all commutator segments when the motor has accelerated to approximately half the speed. This short-circuited condition produces the equivalent of a squirrel-cage rotor for normal-speed running.

Such repulsion-induction or repulsion-start induction motors are very much suitable for use in the rural areas that may be served by long, single-phase lines, since their high starting torque is available without excessive current, and their running speed is nearly constant over a wide torque range. However, these motors have now been replaced by two capacitor single-phase induction motor for most applications.

AV.2 | THREE-PHASE COMMUTATOR MOTORS

Principle

The torque-slip relationship of an inductor motor is

$$T = \frac{3I_2^2 R_2}{s\omega_s} \quad (\text{AV.1})$$

For a given torque the slip is controlled by $I_2^2 R_2$, the rotor copper loss or, in more general terms, the electrical power removed from (or injected into) the rotor circuit. This is efficiently achieved by injecting a slip-frequency emf into the rotor circuit as shown in Fig. AV.2. The torque-slip relation of Eq. (AV.1) now generalizes to

$$T = \frac{3(I_2^2 R_2 \pm E_i I_2 \cos \phi_2)}{s \omega_s} \quad (\text{AV.2})$$

$$s \bar{E}_2 \pm \bar{E}_i = \bar{I}_2 \bar{Z}_2$$

$$\text{or} \quad \bar{I}_2 = \frac{s \bar{E}_2 \pm \bar{E}_i}{\bar{Z}_2} \quad (\text{AV.3})$$

When the injected emf E_i is in-phase with the rotor induced emf sE_2 , the electrical power $E_2 I_2 \cos \phi_2$ is removed (recovered) from the rotor circuit and the plus sign applies in Eq. (AV.2). On the other hand, when E_i is in phase opposition to sE_2 , the electrical power $E_2 I_2 \cos \phi_2$ is injected into the rotor circuit and the minus sign applies in Eq. (AV.2). It easily follows from Eq. (AV.2) that for a given torque, assuming rotor current I_2 to remain nearly constant*, as the magnitude of the in-phase emf E_i is increased, the slip increases (speed decreases). When the phase of E_i is reversed and as E_i is increased in magnitude, the numerator in Eq. (AV.2) becomes negative and so does the slip, i.e. a super-synchronous speed is achieved.

If instead the phase of the injected emf is changed with the magnitude of E_i remaining fixed, control over the rotor current power factor and therefore over that of the primary input current is achieved. A leading phase angle of E_i would result in pf improvement. This is illustrated in Fig. AV.3 for E_i leading sE_2 by 90° . To a rough approximation the component of E_i in-phase (or in phase opposition) controls motor speed and the quadrature component controls the motor power factor.

Generation of Slip-Frequency EMF

Since the injected emf must have slip frequency, a power converter of variable frequency (and voltage) is needed which interchanges electrical power with the fixed-frequency mains. This would maintain high overall efficiency of the set.

Presently, this is accomplished by means of thyristor frequency converters. An older arrangement used for this purpose was the commutator. The principle of generalized frequency conversion by means of a commutator is explained below with the help of Fig. AV.4.

A dc-wound armature is running at speed n rpm, while the machine poles are made to rotate in the opposite direction

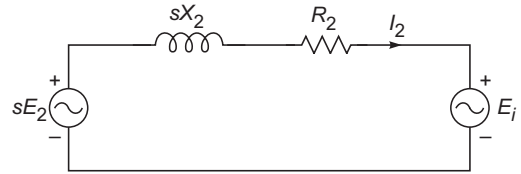


Fig. AV.2 Slip control by injected slip-frequency emf into rotor circuit

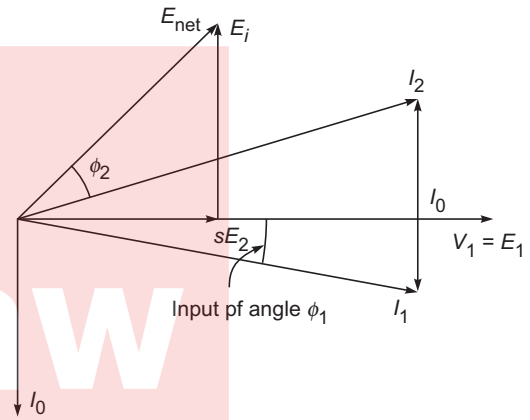


Fig. AV.3 Power factor control by rotor injection

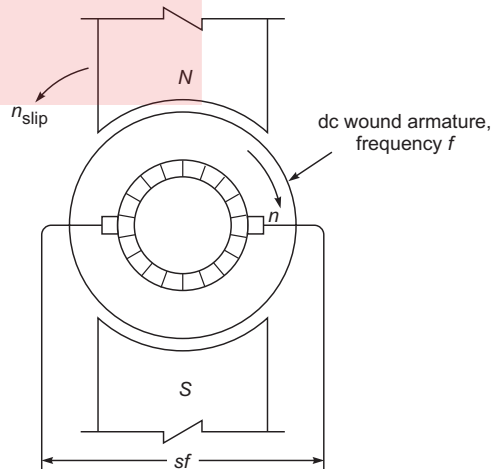


Fig. AV.4 Generalized frequency conversion

* For large values of slip, the rotor power factor ($\cos \phi_2$) decreases and the current for a given torque increases.

at speed n_{slip} rpm with respect to stationary brushes as shown. The frequency of emf induced in the armature (rotor) is

$$f = \frac{(n + n_{\text{slip}})P}{120} = \frac{n_s P}{120} \quad (\text{AV.4})$$

If the poles were stationary, the emf available at the brushes would be dc (zero frequency). For every revolution of poles, the emf of the brushes undergoes one complete alteration (for the 2-pole case illustrated). In general, the frequency of emf at brushes would then be

$$\begin{aligned} f_{\text{brushes}} &= \frac{n_{\text{slip}} P}{120} \\ &= \left(\frac{n_{\text{slip}}}{n_s} \right) f = sf \end{aligned} \quad (\text{AV.5})$$

Thus the frequency of emf in the armature corresponds to the relative speed of poles with respect to the armature, i.e. $n_s = n + n_{\text{slip}}$ and that at brushes corresponds to n_{slip} .

If $n_s = n + n_{\text{slip}}$ is maintained constant, the rms emf per commutator segment would remain constant for fixed flux/pole. The brush emf magnitude could then be varied by tapping a suitable number of commutator segments by symmetrical mechanical adjustment of brushes—moving both brushes by equal angles towards or away from each other. The phase of the brush emf could be varied by moving both the brushes by the same angle in the same direction.

A set of balanced 3-phase emf's of adjustable magnitude and phase could be tapped from the commutator by placing three sets of movable brush pairs as shown in Fig. AV.5.

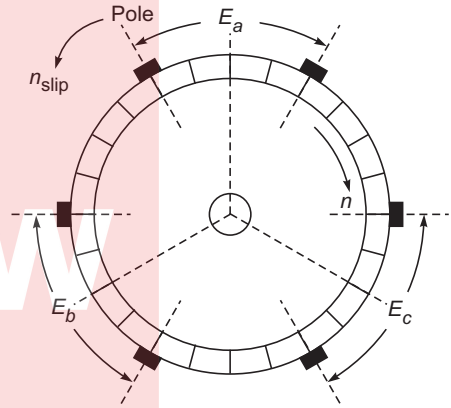


Fig. AV.5 Tapping 3-phase emf's from generalized frequency converter

Schrage Motor

The method explained above of obtaining slip-frequency emf and injecting it into the secondary of an induction motor is incorporated in a single machine called the Schrage motor. It is basically an inverted 3-phase induction motor with the primary winding on the rotor and the secondary winding on the stator. The resultant air-gap field runs at synchronous speed n_s with respect to the rotor and the rotor runs at speed n in opposite direction when the stator winding is shorted. As a consequence, the field runs at slip speed with respect to the stator inducing slip frequency current in it which causes torque production. The rotor also accommodates a dc winding in the same slots as the primary conductors, but placed nearer the slot openings. This arrangement keeps low the leakage inductance of the dc winding coils facilitating commutation. This winding (dc) is called *tertiary* or *regulating winding*. It is connected to a commutator on which are placed three sets of movable brush pairs to collect the 3-phase emf's for injection into the secondary (stator) winding for the control of speed and pf. Since the rotating field moves at slip speed (n_{slip}) with respect to brushes, the frequency of brush emf's is always the slip-frequency. The schematic diagram of a 2-pole Schrage motor is shown in Fig. AV.6.

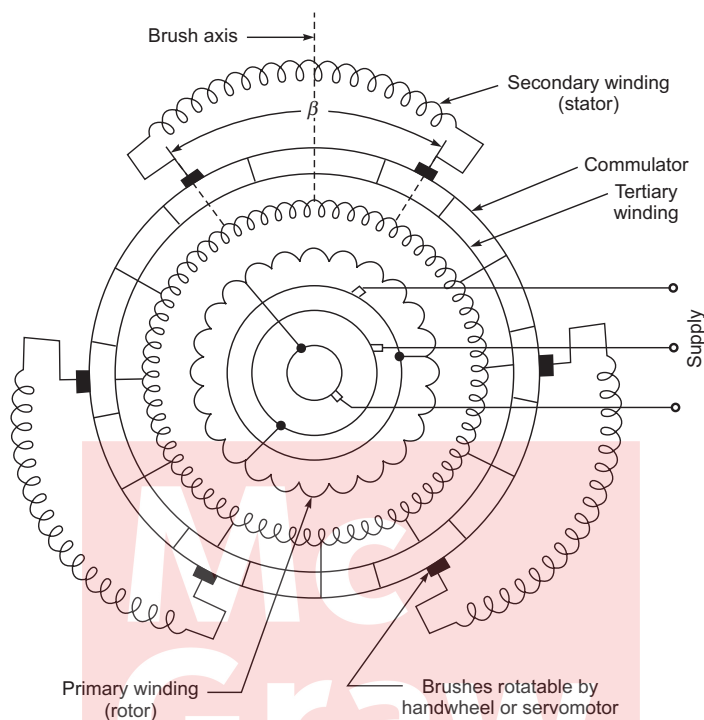


Fig. AV.6 Schematic diagram of a two-pole Schrage motor

By a suitable gear mechanism, the brush pairs can be moved towards or away from each other or both in the same direction giving a mechanical control over the magnitude and phase of the injected emf thereby achieving control over the speed and pf of the motor.

When the brushes are moved so that the injected emf opposes the secondary current, the speed is decreased (positive slip) and for movement in the opposite direction the speed is increased, i.e. the slip is negative. When the brushes are on the same commutator segment, i.e. the brush separation angle is zero, the secondary winding is short-circuited in itself and the motor runs at its natural slip as an inverted induction motor. For an electrical brush separation β , the no-load motor speed is

$$n \approx n_s (1 - C \sin 0.5 \beta) \quad (\text{AV.6})$$

where C is a constant whose value depends upon the turn ratio of the stator and rotor commutator windings and β = elect. brush separation angle. For a given brush separation, the motor power pf can be controlled by moving each brush of brush pairs in the same direction. Starting can be effected by direct switching with brush separation of 180° elect., i.e. one pole-pitch, in the lowest speed position (emf opposing rotor current).

The Schrage motor speed can be varied over the range $0.4n_s$ to $1.4n_s$. The efficiency of the machine on load is slightly lower than that of a corresponding induction motor as the Schrage machine has additional components in the form of commutator and brushes and there are additional losses in the third winding. Commutation problems and design considerations limit the rated output to around 25 kW/pole and placing of the primary winding on the rotor keeps the supply voltage restricted to around 600 V.

The Schrage motor is used as a constant torque drive and the power developed varies directly with speed. It is usually applied in printing presses, textile mills, paper mills, cranes, etc.

Resistance

AVI.1 | CHANGE OF RESISTANCE WITH TEMPERATURE

The dc resistance loss can be found out if the dc resistance is measured. For copper conductors

$$\frac{R_{t1}}{R_{t2}} = \frac{234.5 + t_1}{234.5 + t_2} \quad (\text{AVI.1})$$

where R_{t1} and R_{t2} are the resistances at t_1 and t_2 °C, respectively.

The R_{t1}/R_{t2} versus temperature plot is shown in Fig. AVI.1 for the case $t_2 = 0^\circ\text{C}$.

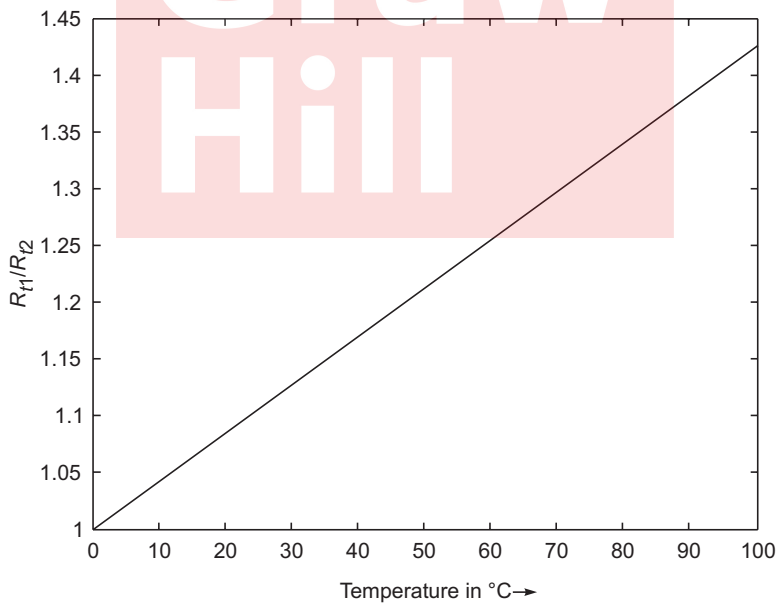


Fig. AVI.1 R_{t1}/R_{t2} versus Temperature Plot

AVI.2 | CHANGE OF RESISTANCE WITH FREQUENCY

ac and dc resistances of copper conductor are related as

$$\frac{R_{AC}}{R_{DC}} = \rho(f) = \text{function of frequency} \quad (\text{AVI.2})$$

When ac is flowing through a conductor, the current is non-uniformly distributed over the cross-section. This effect becomes more pronounced as frequency is increased. This phenomenon is called *skin effect*. It causes larger power loss for a given rms ac than the loss when the same value of dc is flowing through the conductor. Consequently, the effective conductor resistance is more for ac than for dc.

Apart from skin effect, non-uniformity of current distribution is also caused by *proximity effect*. Thus the ratio of R_{AC}/R_{DC} is a function of frequency. R_{AC}/R_{DC} versus frequency plot for copper is shown in Fig. AVI.2.

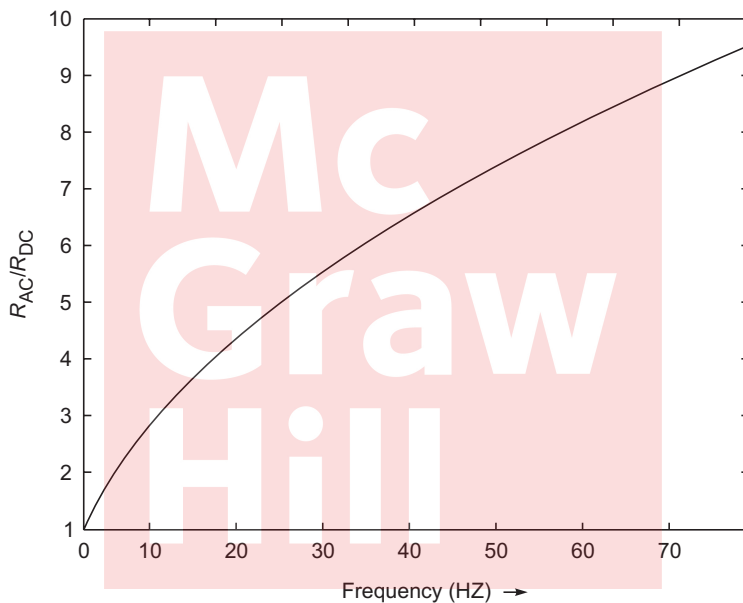


Fig. AVI.2 R_{AC}/R_{DC} versus Frequency Plot

APPENDIX VII

Table of Constants and Unit Conversion

CONSTANTS

Permeability of free space, $\mu_0 = 4\pi \times 10^{-7}$ H/m

Permittivity of free space, $\epsilon_0 = 8.854 \times 10^{-12}$ F/m

Gravitational acceleration, $g = 9.807$ m/s²

UNIT CONVERSION

<i>Quantity</i>	<i>SI Unit</i>	<i>English Unit</i>	<i>CGS Unit</i>
Length	1 m	3.281 ft = 39.37 in	100 cm
Mass	1 kg	2.205 lb (mass); 35.27 ounces (oz)	1000 gm
Force	1 N	0.2248 lb = 7.23 poundals	10 ⁵ dynes
Torque	1 N m	0.738 pound-feet; 141.6 oz.in	10 ⁷ dyne cm
Energy	1 J = 1 Ws = 9.84×10^{-4} BTU = 0.239 Calories	0.738 ft-lb; 2.778×10^{-7} kWh	10 ⁷ ergs
Power	1 W	1.341×10^{-3} hp (1 hp = 746 W) 3.412 BTU/hr	
Moment of inertia	1 kg m ² = 0.738 slug-ft ²	23.73 lb-ft ²	10 ⁷ g cm ² ; 141.6 oz.in sec ²
Magnetic flux	1 Wb	10 ⁸ lines	10 ⁸ maxwells
Magnetic flux density	1 Wb/m ² = 1 T	64.5 kilo lines/in ²	10 ⁴ gauss
Magnetic field intensity	1 A/m	0.254 A/in	0.004 π oersted
Pressure	1 Pa (N/m ²)	4.45×10^{-4} lbf/in ²	9.86×10^{-6} atm

Electric Drives

AVIII.1 | INTRODUCTION TO ELECTRIC DRIVES

To improve manufacturing performance and flexibility, automation control, motion control and machine automation systems are used. Energy consumption can be reduced by engineering assistance with machine safety, energy efficiency, and motion control by braking system automation concepts, global application expertise, and support. Efficient use of new and integrated approaches to complex engineering challenges helps in improved worker's safety. A production machine requires high precision movement, or just simple positioning. This architecture can be implemented using integrated motor-drive technology. It distributes functions such as motion control, safety and predictive maintenance to individual drives.

A modern controller can be used to control the transients and dynamics of the electrical or mechanical machine. Its response is then applied to the loads. It enables self-starting of the motor and can also be used for providing electronic commutation.

AVIII.2 | DRIVES

Drives are defined as a system which is used to employ for motion control. They are basically classified into five categories as follows:

A. On the Basis of Mode of Operation

- (i) **Continuous duty drives:** In these drives, to reach temperature equilibrium, the motor works at a constant load for enough time.
- (ii) **Short time duty drives:** In these drives, the time of operation is very low and the heating time is much lower than the cooling time. The motor operates at constant load and these types of motors are used in crane drives.
- (iii) **Intermittent duty drives:** In these drives, the starting current has little effect on temperature. Temperature equilibrium is never reached by using these drives.

* For detailed discussion please refer to the book on *Electric Drives* by D P Kothari and Rakesh Singh Lodhi published by I K Int New Delhi 2016.

B. On the Basis of Means of Control

- (i) **Manual control drives:** These drives can be either simple like a room fan or complicated like a push button starter.
- (ii) **Semi-automatic control drives:** These drives consist of a manual device for giving a command and automatic device in response to that command.
- (iii) **Automatic control drives:** These drives do not consist of a manual device but have a control gear.

C. On the Basis of Number of Machines

- (i) **Individual drive:** In these drives, a separate motor drives each individual machine.
- (ii) **Group drive:** These drives are also called shaft drives. They consist of one motor, which rotate/control one or more line shafts supported on bearings. The line shaft may be fitted with pulleys, belts, and gears.
- (iii) **Multi-motor drive:** In these drives, a separate motor takes care of each operation of the mechanism. There are several drives in this drive system; it serves to make one of the working parts of drive mechanism error free.

D. On the Basis of Dynamics and Transients

- (i) Independent transient period
- (ii) Dependent transient period

E. On the Basis of Methods of Speed Control

- (i) Reversible and non-reversible independent constant speed
- (ii) Reversible and non-reversible step speed control
- (iii) Variable position control
- (iv) Reversible and non-reversible smooth speed control

AVIII.2 | ELECTRICAL DRIVES

An electrical drive is an electromechanical energy conversion (EMEC) device used for transforming electrical energy into mechanical energy. It is used in various kinds of process control to impart motion to different machines and mechanisms.

An electrical drive system has the following components (Fig. AVIII.1)

1. Electrical machines and loads
2. Motor
3. Power Modulator
4. Sources
5. Control unit
6. Sensing unit

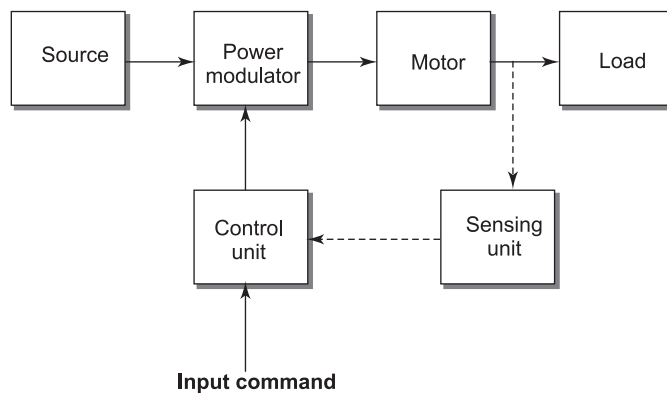


Fig. AVIII.1 Block diagram of electrical drives

Types of Power Modulators

In an electric drive system, the power modulators are of the following types:

1. Controlled rectifiers (ac to dc converters)
2. Inverters (dc to ac converters)
3. ac voltage controllers (ac to ac converters)
4. dc choppers (dc to dc converters)
5. Cycloconverters (Frequency conversion)

Classification of Electrical Drives

The schematic representation of classification of electrical drives is shown in Fig. AVIII.2.

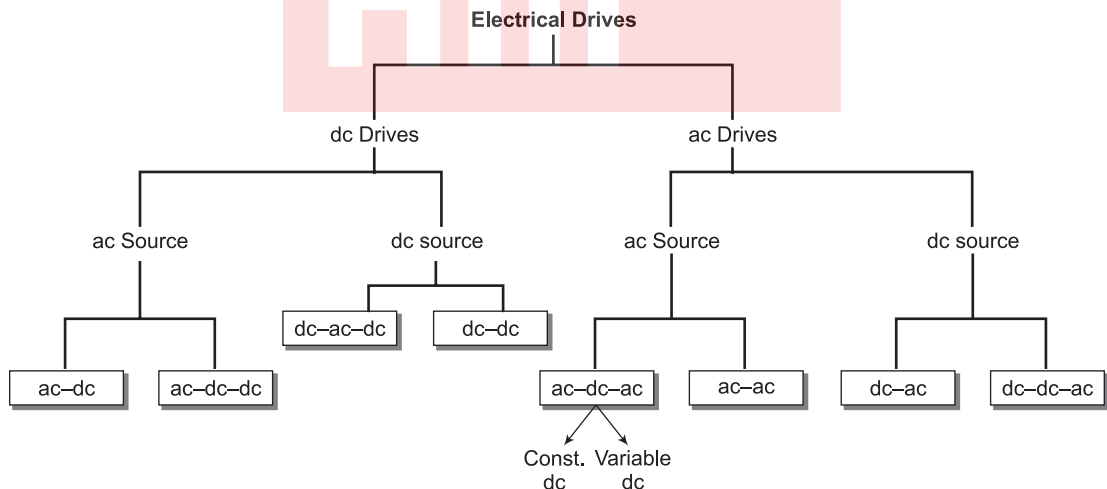


Fig. AVIII.2 Classification of Electrical Drives

Main Parts of a dc Machine

Figure AVIII.3 depicts the main parts of a 4-pole dc machine. They are described as follows:

1. **Stator:** It is the stationary part of the machine. The field winding is placed on stator side. The main parts of a stator are: (i) yoke or frame, (ii) pole core, (iii) pole shoe, (iv) lifting eye, and (v) feet.
2. **Rotor:** It is the rotating or movable part of the machine. The armature winding is placed on rotor side. The main parts of a rotor are: (i) armature core (ii) commutator (iii) shaft, and (iv) armature winding.

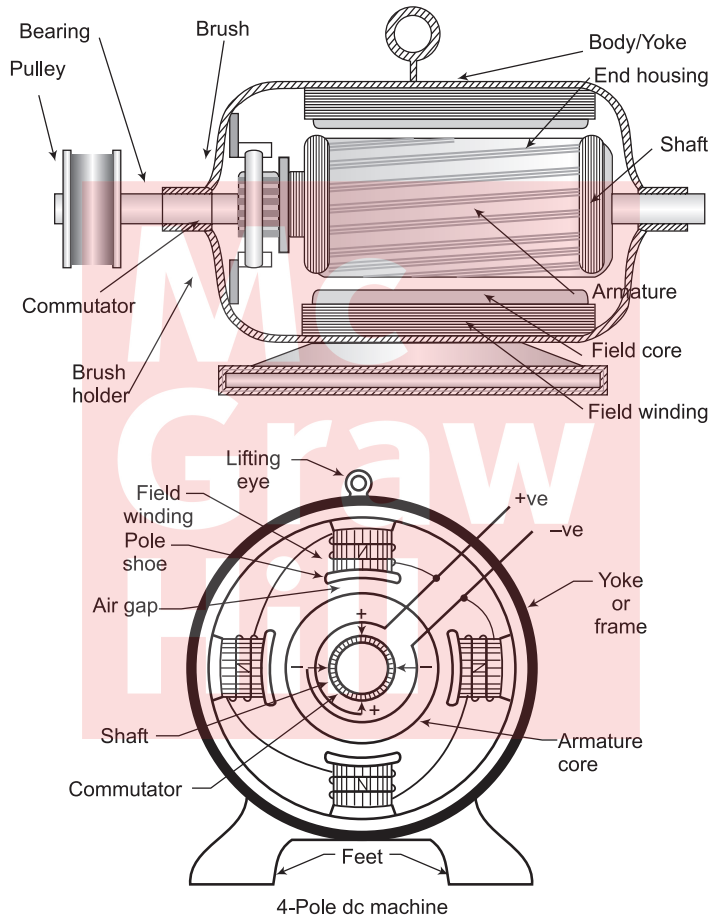


Fig. AVIII.3 Construction of a dc Machine*

Losses in dc Motor

1. **Friction and Windage Losses:** These losses are also known as mechanical losses. These include bearing friction, brush friction, and windage. They vary with changes in speed but remain constant at a given speed. Power losses due to windage increase as the cube of the speed; and those due to friction, increase as the square of the speed.

* For construction of dc machines, refer Chapter 7.

2. **Armature Copper Losses:** These are the I^2R losses of the armature circuit which includes armature winding, commutator and brushes. They vary as the square of currents and directly with the resistance.
3. **Field Copper Losses:** These are I^2R losses of series and shunt field winding, interpole windings and any shunts used in connection with these windings. They vary as the square of currents and directly with the resistance.
4. **Core Losses:** Core losses are basically the eddy current and hysteresis losses in the armature. An expenditure of energy is required with continuous change of direction of flux in the armature iron. These losses are also known as iron losses and magnetic losses. Iron is conducting and moves in a magnetic field so voltage is produced due to these results in small circulating currents called eddy currents. Hysteresis and eddy current losses vary with flux density and speed.

Thyristor Controlled dc Drives

Thyristor controlled drives operate under switching frequency equal to supply frequency according to converter. These drives are employed in low performance, low cost, low power applications as well as high power applications. Figure AVIII.4 shows the circuit of a thyristor controlled dc drive.

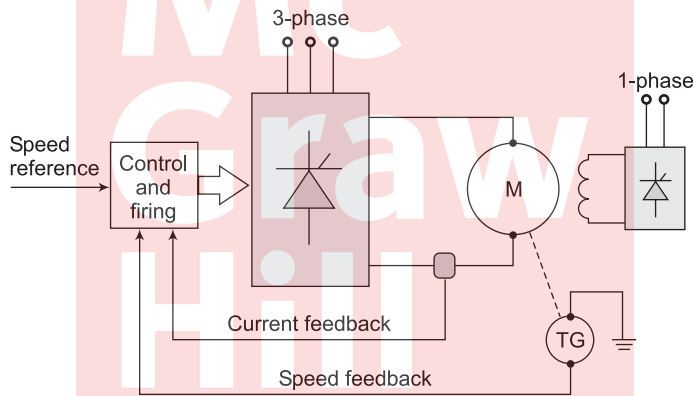


Fig. AVIII.4 Thyristor Controlled dc Drive

Comparison between ac and dc Drives

S. No.	dc Drives	ac Drives
1.	The commutator makes the motor bulky, costly and heavy.	Motors are inexpensive, particularly squirrel cage motor.
2.	The converter technology is well established. The power converter is simple and inexpensive.	The inverter technology is still being developed. The power circuit of the converter and its control are simple.
3.	Line commutation of the converter is used	Forced commutation is used with induction motors. Sometimes machine commutation may be used with synchronous motors.
4.	Fast response and wide speed range smooth control.	With solid state converter, the speed range is wide. With conventional methods, it is stepped and limited.
5.	Small power/weight ratio.	Large power/weight ratio.
6.	Cost does not depend on the solid-state converter	Solid-state converter employed also decides the cost.

Closed Loop Operation of dc Motor

To reach the required velocity, a closed loop controller regulates the power delivered to the motor. If the motor needs to run faster than the required velocity then the controller will deliver less power to the motor.

Advantages of Control Scheme

- It is cost effective.
- It is easy on hardware design.
- It can be retrofitted on systems in which motor does not come along with the encoder.
- Small motors with gear used for hobby robotics usually run at high speed (gear runs at low speed, gear reduction) and can easily use this scheme.

Disadvantages of Control Scheme

- It cannot be used to control motors running at extremely low speeds as motors do not generate sufficient back emf in such cases.
- It cannot be used in system where torque ripple cannot be tolerated. Torque ripple is caused by stopping the PWM for back emf measurement.
- The back emf measurement circuit becomes complicated if the motor is configured to run in both directions using H-bridge (differential measurement).

Feature and Benefits of ac Drives or Variable Frequency Drives

- **Soft start:** Impact on electrical network is reduced; there are no penalties from utility; reduced stress on motor, coupling and load, hence life time extension is given.
- **Precise speed and torque control:** Enhanced product quality; better cost of ownership; improved protection of motor (e.g., stall protection and load); stable product quality despite input power variations and sudden load changes.
- **Wide speed control range:** Better efficiency compared to traditional flow control methods, e.g., damper control and throttling; lower maintenance.
- **High reliability and availability:** Lessened downtime and enhanced process availability.
- **Low audible noise:** Better working environment for operators.
- **Capability for speed reversal/regenerative braking:** Desired torque during braking, therefore better product quality; enhanced braking characteristics; advanced efficiency.
- **Flux optimization (motor flux automatically adapted to load):** Better motor efficiency and reduced motor noise.
- **Power loss ride through:** Diminished number of drive trips and improved process availability.
- **Automatic start (drive can catch a spinning load):** Reduced waiting time and downtime.
- **Energy saving:** ac drives can be retrofired to standard induction motors to provide substantial energy savings.

Braking Methods of Induction Motor Drive

There are three methods of braking in an induction motor. These are as follows:

Generating mode When the energy stored in the rotating mass is dissipated in an electric resistance then the dynamic braking of electric motors occurs. This requires the motor to operate as a generator in order to convert this stored energy into electrical energy.

Regenerative braking mode When the motor speed exceeds the synchronous speed, then occurs the regenerative braking. In regenerative braking, motor works as a generator and electrical power is delivered back to the electrical system. This method of breaking is called regenerative braking.

Plugging The operation shifts from motoring to plugging region, when the phase sequence of supply of motor running at a speed is reversed by interchanging the connections of any two phases of stator, with respect to supply terminals.

Comparison of Induction Motor Starters

Description of Starter	% of line voltage applied	Starting current (I_s) compared with		Starting torque (T_s) compared with	
		D.O.L current (I_{dol})	Full load current (I)	D.O.L Torque (T_{dol})	Full load torque (T)
D.O.L Starter	100	$I_s = I_{dol}$	$I_s = 6 I$	$T_s = T_{dol}$	$T_s = 6 T$
Star delta starter	57.7	$I_s = (1/\sqrt{3})^2 I_{dol}$	$I_s = 2 I$	$T_s = (1/\sqrt{3})^2 T_{dol}$	$T_s = 2/3 T$
Auto transformer starter	80	$I_s = (0.8)^2 I_{dol}$	$I_s = 3.84 I$	$T_s = (0.8)^2 T_{dol}$	$T_s = 1.28 T$
	60	$I_s = (0.6)^2 I_{dol}$	$I_s = 2.16 I$	$T_s = (0.6)^2 T_{dol}$	$T_s = 0.72 T$
	40	$I_s = (0.4)^2 I_{dol}$	$I_s = 0.96 I$	$T_s = (0.4)^2 T_{dol}$	$T_s = 0.32 T$
Reactance-resistance starter	64	$I_s = (0.64)^2 I_{dol}$	$I_s = 2.5 I$	$T_s = (0.425)^2 T_{dol}$	$T_s = 0.35 T$

Speed Control of Induction Motors

Induction motors are of two types, namely—squirrel-cage motor and wound-rotor motor (slip ring rotor motor). Given below are the various types of speed control methods of induction motor.

- Pole changing
- Stator voltage control
- Supply frequency control
- Eddy-current coupling
- Rotor resistance control
- Slip power recovery

Stator Voltage Control

The applications of the stator voltage control method are applicable for torque demand reduced with speed. Examples are fan and pump drives.

Merits of Stator Voltage Control Method

- Simple control circuitry
- Compact size
- Fast response time

Demerits of Stator Voltage Control Method

- Voltage and current waveform gets highly distorted due to harmonics, and therefore affects the efficiency of motor.
- Performance is poor under running condition at low speeds.
- Operating efficiency is low and resistance losses are high.
- Maximum torque of the motor decreases with reduction in stator voltage.

Few Important Electric Drives

1. Static Kramer Drive

An approach has been made to feed the slip energy wastage back to the ac line by converting it into an appropriate signal so as to recover the wastage. Two methods in support of this statement are as follows:

- Static Kramer Drive:** This method is used to obtain an injected voltage that is in phase with the rotor current.
- Static Scherbius Drive:** These drives are used for bi-directional power flow so both positive and negative voltages injected are possible either in phase with or opposing the rotor current.

2. Diesel-electric Drives

These drives use electrical transmission and motor drives to convert movable mechanical power from a diesel engine to traction mechanical power at the train wheels.

Vector Control of Induction Motor

There are two methods of vector control. They are as follows:

1. Direct or feedback method
2. Indirect or feed forward method

Salient Features of Vector Control

- The rotor frequency, ω_e , is not directly controlled as in scalar control. The motor is essentially self-controlled where rotor frequency as well as phase is controlled indirectly with the help of unit vector.
- There is no fear of instability by crossing the operating point beyond the electromagnetic torque, T_{em} , just like in scalar control. Limiting the total I_s within the safe limit automatically limits the operation within stable region.
- Transient response is fast in vector control as dc machines because torque is control by I_{qs} . It does not affect the flux. Signal processing and parameter variation are effected due to delays in converter, so ideal vector control is not possible.
- Speed control is possible in four quadrants as dc machines without using additional control elements. If the torque T_e is negative in forward motoring condition, then the drive initially goes to regenerative braking mode and the speed slows down. At zero speed, phase sequence of the unit vector is reversed, giving reverse motoring operation

Figure AVIII.5 gives a schematic representation of vector control in an induction motor.

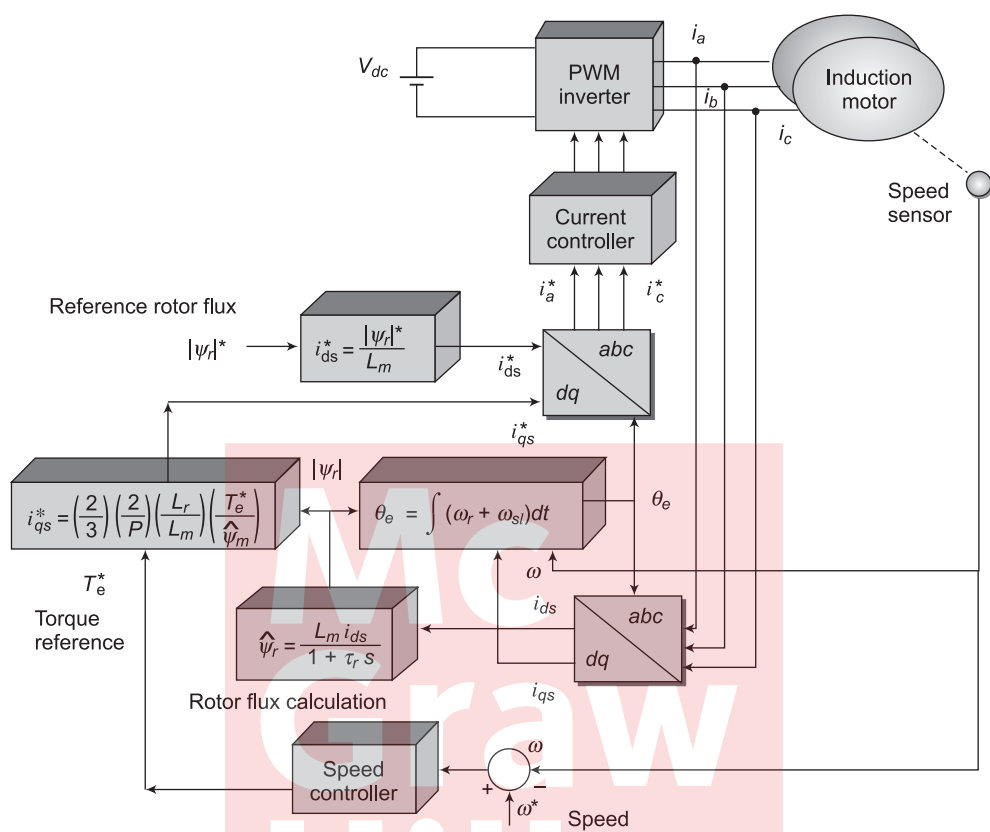


Fig. AVIII.5 Vector control in induction motor